Solutions Manual

Fundamentals of Engineering Electromagnetics

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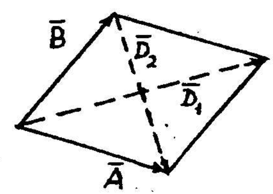
Chapter 2

Vector Analysis

P. 2-1 Denoting the diagonals of the rhombus by $\bar{D}_1$ and $\bar{D}_2$, we have:



(a) $\bar{D}_1 = \bar{A} + \bar{B}$,

$\bar{D}_2 = \bar{A} - \bar{B}$.

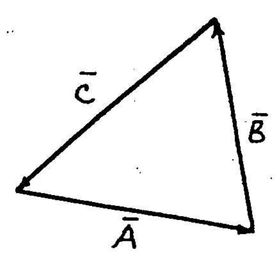
(b) $\bar{D}_1 \cdot \bar{D}_2 = (\bar{A} + \bar{B}) \cdot (\bar{A} - \bar{B})$

$= \bar{A} \cdot \bar{A} - \bar{B} \cdot \bar{B} = 0$,

since $|\bar{A}| = |\bar{B}|$.

Thus, $\bar{D}_1 \perp \bar{D}_2$.

P. 2-2



$$\bar{A} + \bar{B} + \bar{C} = 0.$$

$\bar{A} \times$: $\bar{A} \times \bar{B} = \bar{C} \times \bar{A}$.

$\bar{C} \times$: $\bar{C} \times \bar{A} = \bar{B} \times \bar{C}$.

$\bar{B} \times$: $\bar{B} \times \bar{C} = \bar{A} \times \bar{B}$.

Magnitude relations:

$$AB \sin\theta_{AB} = CA \sin\theta_{CA} = BC \sin\theta_{BC}.$$

Hence,

$$\frac{A}{\sin\theta_{BC}} = \frac{B}{\sin\theta_{CA}} = \frac{C}{\sin\theta_{AB}}. \quad \left(\text{Law of sines.}\right)$$

P. 2-3 a) $\bar{a}_B = \dfrac{\bar{a}_x 4 - \bar{a}_y 6 + \bar{a}_z 12}{\sqrt{4^2 + 6^2 + 12^2}} = \bar{a}_x \dfrac{2}{7} - \bar{a}_y \dfrac{3}{7} + \bar{a}_z \dfrac{6}{7}$.

b) $\bar{B} - \bar{A} = -\bar{a}_x 2 - \bar{a}_y 8 + \bar{a}_z 15$, $\quad |\bar{B} - \bar{A}| = \sqrt{2^2 + 8^2 + 15^2} = 17.1$.

c) $\bar{A} \cdot \bar{a}_B = 6 \times \frac{2}{7} - 2 \times \frac{3}{7} - 3 \times \frac{6}{7} = -1.71$.

d) $\bar{B} \cdot \bar{A} = 24 - 12 - 36 = -24$.

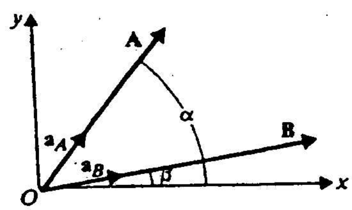
e) $\bar{B} \cdot \bar{a}_A = \dfrac{\bar{B} \cdot \bar{A}}{|\bar{A}|} = \dfrac{-24}{\sqrt{6^2 + 2^2 + 3^2}} = -\dfrac{24}{7} = -3.43$.

f) $\cos\theta_{AB} = \dfrac{\bar{B} \cdot \bar{A}}{BA} = \dfrac{-24}{14 \times 7} = -0.245$, $\quad \theta_{AB} = 180° - 75.8° = 104.2°$

g)
$$\bar{A}\times\bar{C} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ 6 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix} = -\bar{a}_x 4 - \bar{a}_y 3 - \bar{a}_z 10$$

h) $\bar{A}\cdot(\bar{B}\times\bar{C}) = (\bar{A}\times\bar{B})\cdot\bar{C} = -(\bar{A}\times\bar{C})\cdot\bar{B} = -[(-4)4+(-3)(-6)+(-10)12] = -118.$

P. 2-4



$$\bar{a}_A = \bar{a}_x\cos\alpha + \bar{a}_y\sin\alpha,$$
$$\bar{a}_B = \bar{a}_x\cos\beta + \bar{a}_y\sin\beta.$$

a) $\bar{a}_A\cdot\bar{a}_B = \cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$

b)
$$\bar{a}_B\times\bar{a}_A = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \cos\beta & \sin\beta & 0 \\ \cos\alpha & \sin\alpha & 0 \end{vmatrix} = \bar{a}_z(\sin\alpha\cos\beta - \cos\alpha\sin\beta)$$
$$= \bar{a}_z\sin(\alpha-\beta).$$
$$\therefore\ \sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$

P. 2-5 a) $\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = -\bar{a}_x 4 + \bar{a}_y + \bar{a}_z 3,$
$\overrightarrow{P_2P_3} = \overrightarrow{OP_3} - \overrightarrow{OP_2} = \bar{a}_x 6 - \bar{a}_y 5 + \bar{a}_z,$
$\overrightarrow{P_1P_3} = \overrightarrow{OP_3} - \overrightarrow{OP_1} = \bar{a}_x 2 - \bar{a}_y 4 + \bar{a}_z 4.$
$\overrightarrow{P_1P_2}\cdot\overrightarrow{P_1P_3} = 0. \longrightarrow$ Right angle at corner P_1.

b) Area of triangle $= \frac{1}{2}\left|\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}\right| = \frac{1}{2}\left|\overrightarrow{P_1P_2}\right|\left|\overrightarrow{P_1P_3}\right| = 15.3.$

P. 2-6 a) $\overrightarrow{P_1P_2} = \bar{a}_x 2 + \bar{a}_y 4 - \bar{a}_z 4,\quad \left|\overrightarrow{P_1P_2}\right| = \sqrt{2^2+4^2+4^2} = 6.$

b) Perpendicular distance from P_3 to the line
$= \left|\overrightarrow{P_3P_1}\times\bar{a}_{P_1P_2}\right| = \left|(\overrightarrow{OP_1}-\overrightarrow{OP_3})\times\frac{1}{6}\overrightarrow{P_1P_2}\right|$
$= \left|(-\bar{a}_x 5 - \bar{a}_y)\times\frac{1}{6}(\bar{a}_x 2 + \bar{a}_y 4 - \bar{a}_z 4)\right| = \frac{1}{6}\left|\bar{a}_x 4 - \bar{a}_y 20 - \bar{a}_z 18\right| = 4.53.$

P.2-7 Given: $\bar{A} = \bar{a}_x 5 - \bar{a}_y 2 + \bar{a}_z$.

a) Let $\bar{a}_B = \bar{a}_x B_x + \bar{a}_y B_y + \bar{a}_z B_z$,
where $(B_x^2 + B_y^2 + B_z^2)^{1/2} = 1$. (1)

$\bar{a}_B \parallel \bar{A}$ requires $\bar{a}_B \times \bar{A} = 0 = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ B_x & B_y & B_z \\ 5 & -2 & 1 \end{vmatrix}$,

where yields:

$$B_y + 2B_z = 0, \quad (2a)$$
$$-B_x + 5B_z = 0, \quad (2b)$$
$$-2B_x - 5B_y = 0. \quad (2c)$$

Equations (2a), (2b), and (2c) are not all independent. Solving Eqs. (1) and (2), we obtain

$$B_x = \frac{5}{\sqrt{30}}, \quad B_y = -\frac{2}{\sqrt{30}}, \text{ and } B_z = \frac{1}{\sqrt{30}}$$

$$\therefore \quad \bar{a}_B = \frac{1}{\sqrt{30}}(\bar{a}_x 5 - \bar{a}_y 2 + \bar{a}_z).$$

b) Let $\bar{a}_C = \bar{a}_x C_x + \bar{a}_y C_y + \bar{a}_z C_z$, where $C_z = 0$,
and $C_x^2 + C_y^2 = 1$. (3)

$\bar{a}_C \perp \bar{A}$ requires $\bar{a}_C \cdot \bar{A} = 0$, or

$$5C_x - 2C_y = 0. \quad (4)$$

Solution of Eqs. (3) and (4) yields

$$C_x = \frac{2}{\sqrt{29}}, \quad \text{and} \quad C_y = \frac{5}{\sqrt{29}}.$$

$$\therefore \quad \bar{a}_C = \frac{1}{\sqrt{29}}(\bar{a}_x 2 + \bar{a}_y 5).$$

P.2-8 Given: $\bar{A} = \bar{A}_1 + \bar{A}_2 = \bar{a}_x 2 - \bar{a}_y 5 + \bar{a}_z 3$,

$$\bar{B} = -\bar{a}_x + \bar{a}_y 4,$$
$$\bar{A}_1 \perp \bar{B} \longrightarrow \bar{A}_1 \cdot \bar{B} = 0,$$
$$\bar{A}_2 \parallel \bar{B} \longrightarrow \bar{A}_2 \times \bar{B} = 0.$$

Solving, we have

$$\bar{A}_1 = \frac{3}{17}(\bar{a}_x 4 + \bar{a}_y + \bar{a}_z 17) \text{ and } \bar{A}_2 = \frac{22}{17}(\bar{a}_x - \bar{a}_y 4).$$

P.2-10
$$\overrightarrow{OP_1} = -\bar{a}_x - \bar{a}_z 2,$$
$$\overrightarrow{OP_2} = \bar{a}_x(r\cos\phi) + \bar{a}_y(r\sin\phi) + \bar{a}_z z$$
$$= \bar{a}_x\left(-\frac{3}{2}\right) + \bar{a}_y\frac{\sqrt{3}}{2} + \bar{a}_z,$$
$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = -\bar{a}_x\frac{1}{2} + \bar{a}_y\frac{\sqrt{3}}{2} + \bar{a}_z 3, \qquad |\overrightarrow{P_1P_2}| = \sqrt{10}.$$
At $P_1(-1,0,-2)$, $\bar{A}_{P_1} = -\bar{a}_x 2 + \bar{a}_z$.
$$\bar{A}_{P_1}\cdot\bar{a}_{P_1P_2} = \bar{A}_{P_1}\cdot\frac{\overrightarrow{P_1P_2}}{|\overrightarrow{P_1P_2}|} = \frac{4}{\sqrt{10}} = 1.265$$

P.2-11 a)
$$\left.\begin{aligned} x &= r\cos\phi = 3\cos 240^\circ = -\frac{3}{2}, \\ y &= r\sin\phi = 3\sin 240^\circ = -3\sqrt{3}/2, \\ z &= -4 \end{aligned}\right\} \left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}, -4\right)$$
b)
$$\left.\begin{aligned} R &= (r^2+z^2)^{1/2} = (3^2+4^2)^{1/2} = 5, \\ \theta &= \tan^{-1}(r/z) = \tan^{-1}\left(\frac{3}{-4}\right) = 143.1^\circ, \\ \phi &= 4\pi/3 = 240^\circ. \end{aligned}\right\} (5, 143.1^\circ, 240^\circ)$$

P.2-12 a) $-\sin\phi$, b) $\sin\theta\sin\phi$, c) $\cos\theta$,
d) $-\bar{a}_z\cos\phi$, e) $-\bar{a}_\phi\cos\theta$, f) $-\bar{a}_\phi\cos\theta$.

P.2-13 a) In Cartesian coordinates, $\bar{A} = \bar{a}_x A_x + \bar{a}_y A_y + \bar{a}_z A_z$.
$$A_r = \bar{a}_r\cdot\bar{A} = (\bar{a}_r\cdot\bar{a}_x)A_x + (\bar{a}_r\cdot\bar{a}_y)A_y + (\bar{a}_r\cdot\bar{a}_z)A_z$$
$$= A_x\cos\phi_1 + A_y\sin\phi_1$$
b) In spherical coordinates, $\bar{A} = \bar{a}_R A_R + \bar{a}_\theta A_\theta + \bar{a}_\phi A_\phi$.
$$A_r = \bar{a}_r\cdot\bar{A} = (\bar{a}_r\cdot\bar{a}_R)A_R + (\bar{a}_r\cdot\bar{a}_\theta)A_\theta + (\bar{a}_r\cdot\bar{a}_\phi)A_\phi$$
$$= A_R\sin\theta_1 + A_\theta\cos\theta_1$$
$$= \frac{A_R r_1}{\sqrt{r_1^2+z_1^2}} + \frac{A_\theta z_1}{\sqrt{r_1^2+z_1^2}}.$$

P.2-14 a) In Cartesian coordinates, $\bar{E} = \bar{a}_x E_x + \bar{a}_y E_y + \bar{a}_z E_z$.

$$E_\theta = \bar{a}_\theta \cdot \bar{E} = (\bar{a}_\theta \cdot \bar{a}_x) E_x + (\bar{a}_\theta \cdot \bar{a}_y) E_y + (\bar{a}_\theta \cdot \bar{a}_z) E_z$$

$$= E_x \cos\theta_1 \cos\phi_1 + E_y \cos\theta_1 \sin\phi_1 - E_z \sin\theta_1.$$

b) In cylindrical coordinates, $\bar{E} = \bar{a}_r E_r + \bar{a}_\phi E_\phi + \bar{a}_z E_z$.

$$E_\theta = \bar{a}_\theta \cdot \bar{E} = (\bar{a}_\theta \cdot \bar{a}_r) E_r + (\bar{a}_\theta \cdot \bar{a}_\phi) E_\phi + (\bar{a}_\theta \cdot \bar{a}_z) E_z$$

$$= E_r \cos\theta_1 - E_z \sin\theta_1.$$

P.2-15 a) $$\bar{F}_P = \bar{a}_R \frac{12}{\sqrt{(-2)^2+(-4)^2+4^2}} = \bar{a}_R \frac{12}{6} = \bar{a}_R 2.$$

$$(F_P)_y = 2\left(\frac{-4}{\sqrt{(-2)^2+(-4)^2+4^2}}\right) = -\frac{4}{3}$$

b) $$\bar{a}_F = \frac{1}{6}(-\bar{a}_x 2 - \bar{a}_y 4 + \bar{a}_z 4) = \frac{1}{3}(-\bar{a}_x - \bar{a}_y 2 + \bar{a}_z 2).$$

$$\bar{a}_A = \frac{1}{\sqrt{2^2+(-3)^2+(-6)^2}}(\bar{a}_x 2 - \bar{a}_y 3 - \bar{a}_z 6) = \frac{1}{7}(\bar{a}_x 2 - \bar{a}_y 3 - \bar{a}_z 6).$$

$$\theta_{FA} = \cos^{-1}(\bar{a}_F \cdot \bar{a}_A) = \cos^{-1}\frac{1}{21}(-2+6-12) = \cos^{-1}\left(\frac{-8}{21}\right)$$

$$= \cos^{-1}(-0.381) = 180^\circ - 67.6^\circ = 112.4^\circ.$$

P.2-16 $$\int_{P_1}^{P_2} \bar{E} \cdot d\bar{\ell} = \int_{P_1}^{P_2} (y\,dx + x\,dy).$$

a) $x = 2y^2$, $dx = 4y\,dy$; $\displaystyle\int_{P_1}^{P_2} \bar{E} \cdot d\bar{\ell} = \int_1^2 (4y^2 dy + 2y^2 dy) = 14.$

b) $x = 6y - 4$, $dx = 6\,dy$; $\displaystyle\int_{P_1}^{P_2} \bar{E} \cdot d\bar{\ell} = \int_1^2 [6y\,dy + (6y-4)]\,dy = 14.$

Equal line integrals along two specific paths do not necessarily imply a conservative field. $\bar{E}$ is a conservative field in this case because $\bar{E} = \bar{\nabla}(xy + c)$.

P.2-17 a) $\bar{R} = \bar{a}_x x + \bar{a}_y y + \bar{a}_z z$, $\quad \dfrac{1}{R} = (x^2+y^2+z^2)^{-1/2}$.

$$\bar{\nabla}\left(\frac{1}{R}\right) = \bar{a}_x \frac{\partial}{\partial x}\left(\frac{1}{R}\right) + \bar{a}_y \frac{\partial}{\partial y}\left(\frac{1}{R}\right) + \bar{a}_z \frac{\partial}{\partial z}\left(\frac{1}{R}\right)$$

$$= -\frac{1}{R^3}(\bar{a}_x x + \bar{a}_y y + \bar{a}_z z) = -\bar{R}/R^3.$$

b) $\bar{R} = \bar{a}_R R$, $\quad \bar{\nabla}\left(\frac{1}{R}\right) = \bar{a}_R \frac{\partial}{\partial R}\left(\frac{1}{R}\right) = -\bar{a}_R\left(\frac{1}{R^2}\right) = -\bar{R}/R^3.$

P. 2-18 a) $\bar{\nabla}V = \bar{a}_x(2y+z) + \bar{a}_y(2x-z) + \bar{a}_z(x-y)$
$= \bar{a}_x(-2) + \bar{a}_y 4 + \bar{a}_z 3$; Magnitude $= \sqrt{29}$.

b) $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \bar{a}_x(-2) + \bar{a}_y 3 + \bar{a}_z 6$,

$$\bar{a}_{PQ} = \frac{\overrightarrow{PQ}}{\sqrt{(-2)^2+3^2+6^2}} = \frac{1}{7}(-\bar{a}_x 2 + \bar{a}_y 3 + \bar{a}_z 6).$$

Rate of increase of V from P toward $Q = (\bar{\nabla}V)\cdot\bar{a}_{PQ}$
$= \frac{1}{7}(4+12+18) = \frac{34}{7}$.

P. 2-19 a) $\frac{\partial \bar{a}_r}{\partial \phi} = \bar{a}_\phi$; $\frac{\partial \bar{a}_\phi}{\partial \phi} = -\bar{a}_r$.

b)
$$\begin{aligned}
\bar{\nabla}\cdot\bar{A} &= \left(\bar{a}_r\frac{\partial}{\partial r} + \bar{a}_\phi\frac{\partial}{r\partial\phi} + \bar{a}_z\frac{\partial}{\partial z}\right)\cdot(\bar{a}_r A_r + \bar{a}_\phi A_\phi + \bar{a}_z A_z) \\
&= \frac{\partial A_r}{\partial r} + \bar{a}_\phi\frac{1}{r}\cdot\frac{\partial}{\partial\phi}(\bar{a}_r A_r) + \frac{\partial A_\phi}{r\partial\phi} + \frac{\partial A_z}{\partial z} \\
&= \frac{\partial A_r}{\partial r} + \bar{a}_\phi\frac{1}{r}\cdot\left(\bar{a}_r\frac{\partial A_r}{\partial\phi} + A_r\frac{\partial\bar{a}_r}{\partial\phi}\right) + \frac{\partial A_\phi}{r\partial\phi} + \frac{\partial A_z}{\partial z} \\
&= \frac{\partial A_r}{\partial r} + \frac{A_r}{r} + \frac{\partial A_\phi}{r\partial\phi} + \frac{\partial A_z}{\partial z} \\
&= \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{\partial A_\phi}{r\partial\phi} + \frac{\partial A_z}{\partial z}.
\end{aligned}$$

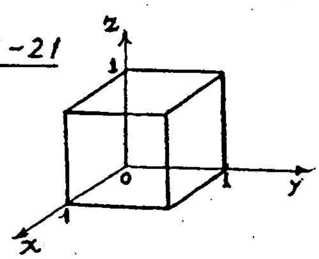
P. 2-20 In spherical coordinates,

$$\bar{\nabla}\cdot\bar{A} = \frac{1}{R^2}\frac{\partial}{\partial R}(R^2 A_R), \quad \text{if} \quad \bar{A} = \bar{a}_R A_R.$$

a) $\bar{A} = f_1(\bar{R}) = \bar{a}_R R^n$, $\quad A_R = R^n$.
$$\bar{\nabla}\cdot\bar{A} = \frac{1}{R^2}\frac{\partial}{\partial R}(R^{n+2}) = (n+2)R^{n-1}.$$

b) $\bar{A} = f_2(\bar{R}) = \bar{a}_R\frac{k}{R^2}$, $\quad A_R = kR^{-2}$.
$$\bar{\nabla}\cdot\bar{A} = \frac{1}{R^2}\frac{\partial}{\partial R}(k) = 0.$$

P. 2-21



$\bar{F} = \bar{a}_x xy + \bar{a}_y yz + \bar{a}_z zx$. To find $\oint \bar{F}\cdot d\bar{s}$:

a) Left face: $y=0$, $d\bar{s} = -\bar{a}_y dx\,dz$.

$$\int_0^1\int_0^1 -yz\,dx\,dz = 0. \qquad (1)$$

Right face: $y=1$, $d\bar{s}=\bar{a}_y\,dx\,dz$.

$$\int_0^1\int_0^1 z\,dx\,dz=\frac{1}{2}. \qquad (2)$$

Top face: $z=1$, $d\bar{s}=\bar{a}_z\,dx\,dy$.

$$\int_0^1\int_0^1 dx\,dy=\frac{1}{2}. \qquad (3)$$

Bottom face: $z=0$, $d\bar{s}=-\bar{a}_z\,dx\,dy$, $\int\bar{F}\cdot d\bar{s}=0$. (4)

Front face: $x=1$, $d\bar{s}=\bar{a}_x\,dy\,dz$.

$$\int_0^1\int_0^1 y\,dy\,dz=\frac{1}{2}. \qquad (5)$$

Back face: $x=0$, $d\bar{s}=-\bar{a}_x\,dy\,dz$, $\int\bar{F}\cdot d\bar{s}=0$. (6)

Adding the results in (1), (2), (3), (4), (5), and (6):

$$\oint\bar{F}\cdot d\bar{s}=\frac{3}{2}.$$

b) $\bar{\nabla}\cdot\bar{F}=y+z+x$, $dv=dx\,dy\,dz$.

$$\int\bar{\nabla}\cdot\bar{F}\,dv=\int_0^1\int_0^1\int_0^1(x+y+z)\,dx\,dy\,dz=\frac{3}{2}.$$

<u>P.2-22</u> $\bar{A}=\bar{a}_r r^2+\bar{a}_z 2z$.

$$\oint_S\bar{A}\cdot d\bar{s}=\left(\int_{\substack{top\\face}}+\int_{\substack{bottom\\face}}+\int_{walls}\right)\bar{A}\cdot d\bar{s}.$$

Top face $(z=4)$: $\bar{A}=\bar{a}_r r^2+\bar{a}_z 8$, $d\bar{s}=\bar{a}_z ds$.

$$\int_{\substack{top\\face}}\bar{A}\cdot d\bar{s}=\int_{\substack{top\\face}}8\,ds=8(\pi 5^2)=200\pi.$$

Bottom face $(z=0)$: $\bar{A}=\bar{a}_r r^2$, $d\bar{s}=-\bar{a}_z ds$, $\int_{\substack{bottom\\face}}\bar{A}\cdot d\bar{s}=0$.

Walls $(r=5)$: $\bar{A}=\bar{a}_r 25+\bar{a}_z 2z$, $d\bar{s}=\bar{a}_r ds$.

$$\int_{Walls}\bar{A}\cdot d\bar{s}=25\int_{Walls}ds=25(2\pi 5\times 4)=1000\pi.$$

$$\therefore\ \oint\bar{A}\cdot d\bar{s}=200\pi+0+1000\pi=1{,}200\pi.$$

$$\bar{\nabla}\cdot\bar{A}=3r+2,\quad \int_V\bar{\nabla}\cdot\bar{A}\,dv=\int_0^4\int_0^{2\pi}\int_0^5(3r+2)r\,dr\,d\phi\,dz=1{,}200\pi.$$
$$=\oint\bar{A}\cdot d\bar{s}.$$

P. 2-23 $\bar{A} = \bar{a}_z z = \bar{a}_z R\cos\theta$.

a) Over the hemispherical surface: $d\bar{s} = \bar{a}_R R^2 \sin\theta\, d\theta\, d\phi$.

$$\int \bar{A}\cdot d\bar{s} = \int_0^{\pi/2}\int_0^{2\pi} \bar{a}_z (R\cos\theta)\cdot \bar{a}_R R^2 \sin\theta\, d\theta\, d\phi$$

$$= R^3\, 2\pi \int_0^{\pi/2} \cos^2\theta \sin\theta\, d\theta = \frac{2}{3}\pi R^3.$$

Over the flat base: $z=0$, $\bar{A}=0$, $\int \bar{A}\cdot d\bar{s} = 0$.

$$\therefore \oint \bar{A}\cdot d\bar{s} = \frac{2}{3}\pi R^3.$$

b) $\bar{\nabla}\cdot\bar{A} = \dfrac{\partial A_z}{\partial z} = \dfrac{\partial z}{\partial z} = 1$.

c) $\int \bar{\nabla}\cdot\bar{A}\, dv = 1\times$ (volume of hemispherical region) $= \frac{2}{3}\pi R^3$

$= \oint \bar{A}\cdot d\bar{s} \longrightarrow$ Divergence theorem is proved.

P. 2-24 $\bar{D} = \bar{a}_R \dfrac{\cos^2\phi}{R^3}$. $\quad d\bar{s} = \begin{cases} \bar{a}_R R^2 \sin\theta\, d\theta\, d\phi, & \text{at } R=3. \\ -\bar{a}_R R^2 \sin\theta\, d\theta\, d\phi, & \text{at } R=2. \end{cases}$

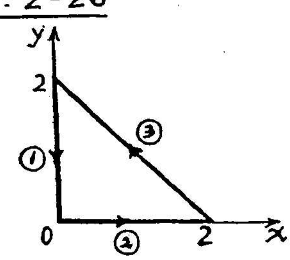
a)
$$\oint \bar{D}\cdot d\bar{s} = \int_0^{2\pi}\int_0^{\pi} \left(\frac{1}{3}-\frac{1}{2}\right)\sin\theta\, d\theta\cdot \cos^2\phi\, d\phi$$

$$= -\frac{1}{6}\int_0^{\pi}\sin\theta\, d\theta \int_0^{2\pi}\cos^2\phi\, d\phi = -\frac{1}{6}(2)\pi = -\frac{\pi}{3}.$$

b) $\bar{\nabla}\cdot\bar{D} = -\dfrac{\cos^2\phi}{R^4}$, $\quad dv = R^2 \sin\theta\, dR\, d\theta\, d\phi$.

$$\int \bar{\nabla}\cdot\bar{D}\, dv = \int_0^{2\pi}\int_0^{\pi}\int_2^{3}\left(-\frac{\cos^2\phi}{R^2}\right)\sin\theta\, dR\, d\theta\, d\phi = -\frac{\pi}{3}.$$

P. 2-26



a) $d\bar{\ell} = \bar{a}_x dx + \bar{a}_y dy$,

$\bar{A}\cdot d\bar{\ell} = (2x^2+y^2)dx + (xy-y^2)dy$.

Path ①: $x=0$, $dx=0$, $\int \bar{A}\cdot d\bar{\ell} = -\int_2^0 y^2 dy = 8/3$.

Path ②: $y=0$, $dy=0$, $\int \bar{A}\cdot d\bar{\ell} = \int_0^2 2x^2 dx = 16/3$.

Path ③: $y=2-x$, $dy=-dx$, $\int \bar{A}\cdot d\bar{\ell} = -28/3$.

$$\oint \bar{A}\cdot d\bar{\ell} = \frac{8}{3} + \frac{16}{3} - \frac{28}{3} = -\frac{4}{3}.$$

b) $\bar{\nabla}\times\bar{A} = -\bar{a}_z y$, $d\bar{s} = \bar{a}_z dx\,dy$, $\int(\bar{\nabla}\times\bar{A})\cdot d\bar{s} = -\int_0^2\left[\int_0^{2-x} y\, dy\right]dx = -\frac{4}{3}$.

c) No. $\bar{\nabla}\times\bar{A} \neq 0$.

P.2-27 $\bar{F} = \bar{a}_r 5r\sin\phi + \bar{a}_\phi r^2\cos\phi$.

a) Path AB: $r=1$, $\bar{F} = \bar{a}_r 5\sin\phi + \bar{a}_\phi \cos\phi$; $d\bar{\ell} = \bar{a}_\phi d\phi$.

$$\int_{AB} \bar{F}\cdot d\bar{\ell} = \int_0^{\pi/2} \cos\phi \, d\phi = 1.$$

Path BC: $\phi = \pi/2$, $\bar{F} = \bar{a}_r 5r$; $d\bar{\ell} = \bar{a}_r dr$.

$$\int_{BC} \bar{F}\cdot d\bar{\ell} = \int_1^2 5r\,dr = 15/2.$$

Path CD: $r=2$, $\bar{F} = \bar{a}_r 10\sin\phi + \bar{a}_\phi 4\cos\phi$; $d\bar{\ell} = \bar{a}_\phi 2 d\phi$.

$$\int_{CD} \bar{F}\cdot d\bar{\ell} = \int_{\pi/2}^{0} 8\cos\phi\, d\phi = -8.$$

Path DA: $\phi = 0$, $\bar{F} = \bar{a}_\phi r^2$; $d\bar{\ell} = \bar{a}_r dr$.

$$\int_{DA} \bar{F}\cdot d\bar{\ell} = 0.$$

$$\therefore \oint_{ABCDA} \bar{F}\cdot d\bar{\ell} = 1 + \frac{15}{2} - 8 = \frac{1}{2}.$$

b) $$\bar{\nabla}\times\bar{F} = \bar{a}_z \frac{1}{r}\left[\frac{\partial}{\partial r}(rF_\phi) - \frac{\partial F_r}{\partial\phi}\right] = \bar{a}_z (3r-5)\cos\phi.$$

c) $d\bar{s} = -\bar{a}_z r\,dr\,d\phi$, $(\bar{\nabla}\times\bar{F})\cdot d\bar{s} = -r(3r-5)dr\cos\phi\, d\phi$.

$$\int (\bar{\nabla}\times\bar{F})\cdot d\bar{s} = -\int_1^2 r(3r-5)dr \int_0^{\pi/2}\cos\phi\, d\phi = \frac{1}{2}.$$

P.2-28 $\bar{A} = \bar{a}_\phi 3\sin(\phi/2)$.

$$\bar{\nabla}\times\bar{A} = \frac{3}{R\sin\theta}\left(\bar{a}_R \cos\theta\sin\frac{\phi}{2} - \bar{a}_\theta \sin\theta\sin\frac{\phi}{2}\right).$$

Assume the hemispherical bowl to be located in the lower half of the xy-plane and its circular rim coincident with the xy-plane. Tracing the rim in a counterclockwise direction, we have

$$d\bar{\ell} = \bar{a}_\phi 4\, d\phi, \quad d\bar{s} = -\bar{a}_R 4^2 \sin\theta\, d\theta\, d\phi.$$

$$\oint_C \bar{A}\cdot d\bar{\ell} = \int_0^{2\pi} (\bar{A})_{\substack{R=4\\ \theta=\pi/2}} \cdot (\bar{a}_\phi 4\, d\phi) = \int_0^{2\pi} 12\sin\left(\frac{\phi}{2}\right) d\phi = 48.$$

$$\int_S (\bar{\nabla}\times\bar{A})\cdot d\bar{s} = -12\int_0^{2\pi}\int_{\pi/2}^{\pi} \cos\theta\sin\frac{\phi}{2}\, d\theta\, d\phi = 48.$$

$$= \oint_C \bar{A}\cdot d\bar{\ell}.$$

P. 2-30. $\bar{F} = \bar{a}_x(x+3y-c_1z) + \bar{a}_y(c_2x+5z) + \bar{a}_z(2x-c_3y+c_4z)$.

a) $\bar{F}$ is irrotational:

$$\bar{\nabla}\times\bar{F} = \bar{a}_x\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \bar{a}_y\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \bar{a}_z\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right) = 0.$$

Each component must vanish.

$$\frac{\partial}{\partial y}(2x - c_3y + c_4z) - \frac{\partial}{\partial z}(c_2x+5z) = 0 \longrightarrow c_3 = -5.$$

$$\frac{\partial}{\partial z}(x+3y-c_1z) - \frac{\partial}{\partial x}(2x - c_3y + c_4z) = 0 \longrightarrow c_1 = -2,$$

$$\frac{\partial}{\partial x}(c_2x+5z) - \frac{\partial}{\partial y}(x+3y-c_1z) = 0 \longrightarrow c_2 = 3.$$

b) F is also solenoidal:

$$\bar{\nabla}\cdot\bar{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 0.$$

$$\frac{\partial}{\partial x}(x+3y-c_1z) + \frac{\partial}{\partial y}(c_2x+5z) + \frac{\partial}{\partial z}(2x-c_3y+c_4z) = 0.$$

$$\longrightarrow c_4 = -1.$$

Chapter 3

Static Electric Fields

P. 3-1 a) Max. voltage V_{max} will make $d_1 = h/2$ at $z = w$.

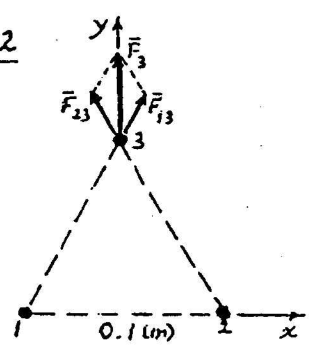
$$\frac{h}{2} = \frac{e}{2m}\left(\frac{V_{max}}{h}\right)\left(\frac{w}{u_0}\right)^2 \longrightarrow V_{max} = \frac{m}{e}\left(\frac{u_0 h}{w}\right)^2.$$

b) At the screen, $(d_0)_{max} = D/2$. L must be $\leq L_{max}$, where

$$L_{max} = \frac{1}{2}\left(w + \frac{m u_0^2 D h}{e w V_{max}}\right).$$

c) Double V_{max} by doubling u_0^2, or doubling the anode accelerating voltage.

P. 3-2



$$\bar{F}_{13} = \frac{(2\times10^{-6})^2}{4\pi\epsilon_0\,(0.1)^2}(\bar{a}_x 0.5 + \bar{a}_y 0.866)$$

$$= 3.6\,(\bar{a}_x 0.5 + \bar{a}_y 0.866)\ \ (N).$$

$$\bar{F}_{23} = 3.6\,(-\bar{a}_x 0.5 + \bar{a}_y 0.866)\ \ (N).$$

$$\bar{F}_3 = \bar{F}_{13} + \bar{F}_{23} = \bar{a}_y\, 0.624\ \ (N).$$

Similarly for $\bar{F}_1$ and $\bar{F}_2$. All are repulsive forces in the direction away from the center of the triangle.

P. 3-3 $\overrightarrow{Q_1P} = -\bar{a}_y 3 + \bar{a}_z 4$, $\quad \overrightarrow{Q_2P} = \bar{a}_y 4 - \bar{a}_z 3$.

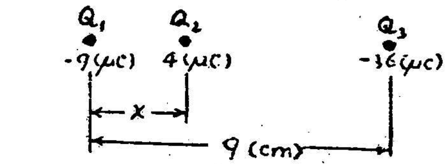
At P: $\bar{E}_1 = \dfrac{Q_1}{4\pi\epsilon_0 (5)^3}(-\bar{a}_y 3 + \bar{a}_z 4)$,

$$\bar{E}_2 = \frac{Q_2}{4\pi\epsilon_0 (5)^3}(\bar{a}_y 4 - \bar{a}_z 3).$$

a) No y-component: $-3Q_1 + 4Q_2 = 0 \longrightarrow \dfrac{Q_1}{Q_2} = \dfrac{4}{3}$.

b) No z-component: $4Q_1 - 3Q_2 = 0 \longrightarrow \dfrac{Q_1}{Q_2} = \dfrac{3}{4}$.

P. 3-4



For zero force on Q_1:

$$\frac{Q_1 Q_2}{4\pi\epsilon_0 x^2} + \frac{Q_1 Q_3}{4\pi\epsilon_0 9^2} = 0.$$

$$x = 9\sqrt{\frac{Q_2}{Q_3}} = 9\sqrt{\frac{4}{36}} = 3 \text{ (cm)}.$$

With $x = 3$ (cm), it can be proved that the net forces on Q_2 and Q_3 are also zero.

P. 3-5 From Eq. (3-42a), $\rho_s = \frac{\text{Total charge}}{\text{Disk area}} = \frac{Q}{\pi b^2}$.

$$\bar{E} = \bar{a}_z \frac{\rho_s}{2\epsilon_0}\left[1-\left(1+\frac{b^2}{z^2}\right)^{-1/2}\right]$$

$$= \bar{a}_z \frac{\rho_s}{2\epsilon_0}\left[1-\left(1-\frac{b^2}{2z^2}+\frac{3}{8}\frac{b^4}{z^4}-\cdots\right)\right]$$

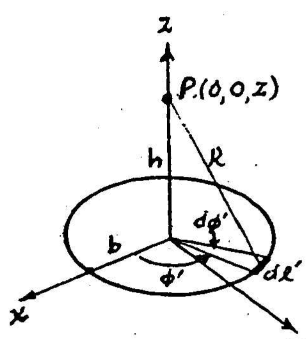
$$= \bar{a}_z \frac{Q}{2\pi\epsilon_0 b^2}\left[\frac{1}{2}\left(\frac{b}{z}\right)^2-\frac{3}{8}\left(\frac{b}{z}\right)^4+\cdots\right]$$

$$= \bar{a}_z\left[\frac{Q}{4\pi\epsilon_0 z^2}\left(1-\frac{3}{4}\frac{b^2}{z^2}+\cdots\right)\right],$$

where the first term is the point-charge term and the rest represent the error. Considering only the first error term:

$$\frac{3}{4}\cdot\frac{b^2}{z^2} \le 0.01. \longrightarrow z \ge \sqrt{75}\, b, \text{ or } 8.66\, b.$$

P. 3-6 At an arbitrary $P(0,0,z)$ on the axis:



$$dV_P = \frac{\rho_\ell b\, d\phi'}{4\pi\epsilon_0 (z^2+b^2)^{1/2}}.$$

$$V_P = \frac{\rho_\ell b}{4\pi\epsilon_0 (z^2+b^2)^{1/2}}\int_0^{2\pi} d\phi' = \frac{\rho_\ell b}{2\epsilon_0 (z^2+b^2)^{1/2}}.$$

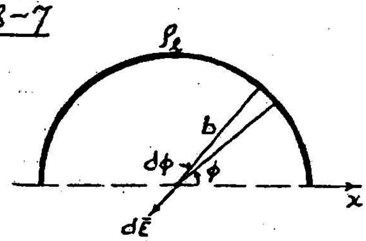
$$\bar{E}_P = -\bar{\nabla} V_P = -\bar{a}_z \frac{dV_P}{dz} = \bar{a}_z \frac{\rho_\ell b}{2\epsilon_0 (z^2+b^2)^{3/2}}.$$

a) At point $(0,0,h)$, $\bar{E} = \bar{a}_z \frac{\rho_\ell b}{2\epsilon_0 (h^2+b^2)^{3/2}}$.

b) To find the location of max. $|\bar{E}_P|$, set $\frac{\partial}{\partial z}|\bar{E}_P| = 0 \longrightarrow z = \frac{b}{\sqrt{2}}$. $\text{Max}|\bar{E}_P| = \frac{\rho_\ell}{3.67\epsilon_0 b^2}$.

Similar situation when P is below the loop.

P.3-7



$$dE_y = -\frac{\rho_\ell (b\,d\phi)}{4\pi\epsilon_0 b^2}\sin\phi,$$

$$\bar{E} = \bar{a}_y E_y = -\bar{a}_y \frac{\rho_\ell}{4\pi\epsilon_0 b}\int_0^\pi \sin\phi\, d\phi$$

$$= -\bar{a}_y \frac{\rho_\ell}{2\pi\epsilon_0 b}.$$

P.3-8 Spherical symmetry: $\bar{E} = \bar{a}_R E_R$. Apply Gauss's law.

1) $0 \leq R \leq b$. $4\pi R^2 E_{R1} = \frac{\rho_0}{\epsilon_0}\int_0^R \left(1 - \frac{R^2}{b^2}\right) 4\pi R^2 dR = \frac{4\pi\rho_0}{\epsilon_0}\left(\frac{R^3}{3} - \frac{R^5}{5b^2}\right),$

$$E_{R1} = \frac{\rho_0}{\epsilon_0} R\left(\frac{1}{3} - \frac{R^2}{5b^2}\right).$$

2) $b \geq R < R_i$. $4\pi R^2 = \frac{\rho_0}{\epsilon_0}\int_0^b \left(1 - \frac{R^2}{b^2}\right) 4\pi R^2 dR = \frac{8\pi\rho_0}{15\epsilon_0} b^3,$

$$E_{R2} = \frac{2\rho_0 b^3}{15\epsilon_0 R^2}.$$

3) $R_i < R < R_o$. $E_{R3} = 0$.

4) $R > R_o$. $E_{R4} = \frac{2\rho_0 b^3}{15\epsilon_0 R^2}$.

P.3-9 Cylindrical symmetry: $\bar{E} = \bar{a}_r E_r$. Apply Gauss's law.

a) $E_r = 0$, for $r < a$.

$E_r = \frac{a\rho_{sa}}{\epsilon_0 r}$, for $a < r < b$.

$E_r = \frac{a\rho_{sa} + b\rho_{sb}}{\epsilon_0 r}$, for $r > b$.

b) $\frac{b}{a} = -\frac{\rho_{sa}}{\rho_{sb}}$.

P.3-10 $W_e = -q\int \bar{E}\cdot d\bar{\ell} = -q\int (y\,dx + x\,dy)$.

a) Along the parabola $y = 2x^2$, $dy = 4x\,dx$.

$$W_e = -(5\times 10^{-6})\int_1^{-2} (2x^2 + 4x^2)\,dx = 9\times 10^{-5}\,(J) = 90\,(\mu J).$$

b) Along the straight line $\frac{y-2}{x-1} = \frac{8-2}{-2-1} = -2$, $y = -2x+4$, $dy = -2dx$.

$$W_e = -(5\times 10^{-6})\int_1^{-2} [(-2x+4)\,dx - 2x\,dx] = 90\,(\mu J).$$

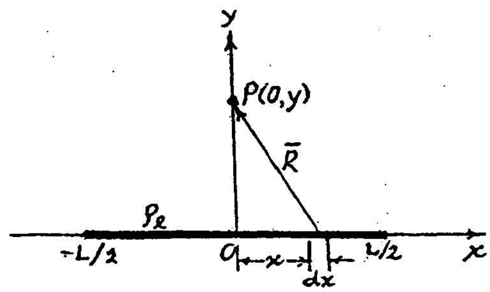
P.3-11 $\bar{E} = \bar{a}_x y - \bar{a}_y x$, $\bar{E}\cdot d\bar{l} = y\,dx - x\,dy.$

a) $W_e = -q\int_1^{-2}(2x^2 - 4x^2)\,dx = -30\ (\mu J).$

b) $W_e = -q\int_1^{-2}[(-2x+4)+2x]\,dx = -60\ (\mu J).$

The given $\bar{E}$ field is nonconservative.

P.3-12



a) $V = 2\int_0^{L/2}\frac{\rho_\ell\,dx}{4\pi\epsilon_0 R}$

$= \frac{\rho_\ell}{2\pi\epsilon_0}\int_0^{L/2}\frac{dx}{\sqrt{x^2+y^2}}$

$= \frac{\rho_\ell}{2\pi\epsilon_0}\left\{\ln\left[\sqrt{\left(\frac{L}{2}\right)^2+y^2}+\frac{L}{2}\right] - \ln y\right\}.$

b) From Coulomb's law:

$$\bar{E} = \bar{a}_y E_y = 2\int_0^{L/2}\bar{a}_y\frac{\rho_\ell\, y\,dx}{4\pi\epsilon_0 R^3} = \bar{a}_y\frac{\rho_\ell}{2\pi\epsilon_0 y}\left[\frac{L/2}{\sqrt{(L/2)^2+y^2}}\right].$$

c) $\bar{E} = -\bar{\nabla}V$ gives the same answer as in b).

P.3-13 a) $\rho_{ps} = \bar{P}\cdot\bar{a}_n = P_0\frac{L}{2}$ on all six faces of the cube.

$\rho_{pv} = -\bar{\nabla}\cdot\bar{P} = -3P_0.$

b) $Q_s = (6L^2)\rho_{ps} = 3P_0L^3$; $Q_v = (L^3)\rho_{pv} = -3P_0L^3.$

Total bound charge $= Q_s + Q_v = 0.$

P.3-14 $\bar{P} = \bar{a}_x P_0.$

a) $\rho_{ps} = \bar{P}\cdot\bar{a}_R = P_0\sin\theta\cos\phi.$

$\rho_{pv} = -\bar{\nabla}\cdot\bar{P} = 0.$

b) $Q_s = \int_0^{\pi}\int_0^{2\pi}P_0\, b^2\sin^2\theta\cos\phi\,d\phi\,d\theta$

$= 0.$

P. 3-15 $\bar{P} = P_0(\bar{a}_x 3x + \bar{a}_y 4y)$.

a) $\rho_{pv} = -\bar{\nabla}\cdot\bar{P} = -7P_0$.

Total volume charge $Q_v = -7P_0\pi(r_o^2 - r_i^2)$ per unit length.

Outer surface: $r = r_o$, $\bar{a}_n = \bar{a}_r$. $\rho_{ps_o} = \bar{P}\cdot\bar{a}_r = P_0(\bar{a}_x 3r_o\cos\phi + \bar{a}_y 4r_o\sin\phi)\cdot\bar{a}_r$

$$= P_0 r_o(3\cos^2\phi + 4\sin^2\phi)$$
$$= P_0 r_o(3 + \sin^2\phi).$$

Inner surface: $r = r_i$, $\bar{a}_n = -\bar{a}_r$. $\rho_{ps_i} = -P_0 r_i(3 + \sin^2\phi)$.

b) Total $Q_{so} = \int_0^{2\pi}\rho_{ps_o} r_o\,d\phi = P_0 r_o^2\int_0^{2\pi}(3+\sin^2\phi)\,d\phi = 7\pi P_0 r_o^2$, per unit length.

Total $Q_{si} = -7\pi P_0 r_i^2$, per unit length.

Total bound charge: $Q_v + Q_{so} + Q_{si} = 0$.

P. 3-16 Spherical symmetry: Apply Gauss's law. $\bar{E} = \bar{a}_R E_R$, $\bar{D} = \bar{a}_R D_R$.

(1) $R > R_o$.
$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2}, \qquad V_1 = \frac{Q}{4\pi\epsilon_0 R}.$$
$$D_{R1} = \epsilon_0 E_{R1} = \frac{Q}{4\pi R^2}, \qquad P_{R1} = 0.$$

(2) $R_i < R < R_o$.
$$E_{R2} = \frac{Q}{4\pi\epsilon_0\epsilon_r R^2}, \qquad D_{R2} = \frac{Q}{4\pi R^2},$$
$$P_{R2} = D_{R2} - \epsilon_0 E_{R2} = \left(1 - \frac{1}{\epsilon_r}\right)\frac{Q}{4\pi R^2}.$$
$$V_2 = -\int_\infty^{R_o} E_{R1}\,dR - \int_{R_o}^{R} E_{R2}\,dR = \frac{Q}{4\pi\epsilon_0}\left[\left(1-\frac{1}{\epsilon_r}\right)\frac{1}{R_o} - \frac{1}{\epsilon_r R}\right].$$

(3) $R < R_i$
$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2}, \quad D_{R3} = \frac{Q}{4\pi R^2}, \quad P_{R3} = 0.$$
$$V_3 = V_2\Big|_{R=R_i} - \int_{R_i}^{R} E_{R3}\,dR$$
$$= \frac{Q}{4\pi\epsilon_0}\left[\left(1-\frac{1}{\epsilon_r}\right)\frac{1}{R_o} - \left(1-\frac{1}{\epsilon_r}\right)\frac{1}{R_i} + \frac{1}{R}\right].$$

P.3-17 Use subscript a for air, p for plexiglass, and b for breakdown.

a) $V_b = E_{ba} d_a = (3\times10^6)\times(50\times10^{-3}) = 150\times10^3 (V) = 150\ (kV)$.

b) $V_b = E_{bp} d_p = 20\times50 = 1{,}000\ (kV)$.

c) $V_b = E_a d_a + E_p d_p = E_a(50-d_p) + E_p d_p$

Now $D_a = D_p \longrightarrow \epsilon_0 E_a = \epsilon_0 \epsilon_{rp} E_p \longrightarrow E_a = \epsilon_{rp} E_p > E_p$.

$E_{ba} < E_{bp} \longrightarrow$ Breakdown occurs in air region first.

$\therefore\ V_b = E_{ba}(50-10) + \frac{E_{ba}}{3}\times10 = 3(40-\frac{1}{3}\times10) = 130 (kV)$.

P.3-18 At the z=0 plane: $\bar{E}_1 = \bar{a}_x 2y - \bar{a}_y 3x + \bar{a}_z 5$.

$\bar{E}_{1t}(z=0) = \bar{E}_{2t}(z=0) = \bar{a}_x 2y - \bar{a}_y 3x$.

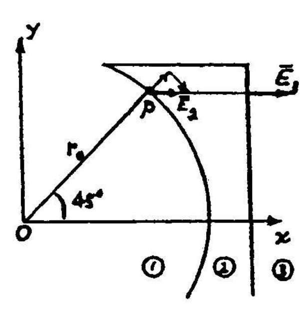
$\bar{D}_{1n}(z=0) = \bar{D}_{2n}(z=0) \longrightarrow 2\bar{E}_{1n}(z=0) = 3\bar{E}_{2n}(z=0)$,

$\longrightarrow \bar{E}_{2n}(z=0) = \frac{2}{3}(\bar{a}_z 5) = \bar{a}_z \frac{10}{3}$.

$\therefore\ \bar{E}_2(z=0) = \bar{a}_x 2y - \bar{a}_y 3x + \bar{a}_z \frac{10}{3}$.

$\bar{D}_2(z=0) = (\bar{a}_x 2y - \bar{a}_y 3x + \bar{a}_z \frac{10}{3}) 3\epsilon_0$

P.3-19



Assume $\bar{E}_2 = \bar{a}_r E_{2r} + \bar{a}_\phi E_{2\phi}$.

Boundary condition: $\bar{a}_n \times \bar{E}_1 = \bar{a}_n \times \bar{E}_2$.

$\longrightarrow E_{2\phi} = -3$.

For $\bar{E}_3$, and hence $\bar{E}_2$, to be parallel to the x-axis, $E_{2\phi} = -E_{2r}$.

$\longrightarrow E_{2r} = 3$.

Boundary condition: $\bar{a}_n \cdot \bar{D}_1 = \bar{a}_n \cdot \bar{D}_2$.

$\longrightarrow \epsilon_1 E_{r1} = \epsilon_2 E_{r2} \longrightarrow \epsilon_0 5 = \epsilon_0 \epsilon_{r2} 3$.

$\longrightarrow \epsilon_{r2} = \frac{5}{3} = 1.667$.

P.3-20 $\epsilon = \frac{\epsilon_2-\epsilon_1}{d}y + \epsilon_1$.

Assume Q on plate at $y=d$. $\bar{E} = -\bar{a}_y \frac{\rho_s}{\epsilon} = \frac{Q}{S(\frac{\epsilon_2-\epsilon_1}{d}y+\epsilon_1)}$.

$$V = -\int_{y=0}^{y=d} \bar{E}\cdot d\bar{\ell} = \frac{Qd \ln(\epsilon_2/\epsilon_1)}{S(\epsilon_2-\epsilon_1)}.$$

$$C = \frac{Q}{V} = \frac{S(\epsilon_2-\epsilon_1)}{d \ln(\epsilon_2/\epsilon_1)}.$$

P.3-21 Let ρ_ℓ be the lineal charge density on the inner conductor. $\bar{E} = \bar{a}_r \frac{\rho_\ell}{2\pi\epsilon r}$.

$$V_0 = -\int_b^a \bar{E}\cdot d\bar{r} = \frac{\rho_\ell}{2\pi\epsilon}\ln\left(\frac{b}{a}\right) \longrightarrow \rho_\ell = \frac{2\pi\epsilon V_0}{\ln(b/a)}.$$

a) $\bar{E}(a) = \bar{a}_r \frac{V_0}{a\ln(b/a)}$.

b) For a fixed b, the function to be minimized is: $(x=b/a)$:
$f(x) = \frac{V_0 x}{b\cdot \ln x}$. Setting $\frac{df(x)}{dx}=0$ yields $\ln x = 1$,
or $x = \frac{b}{a} = e = 2.718$.

c) min. $E(a) = eV_0/b$.

d) $C' = \frac{\rho_\ell}{V_0} = \frac{2\pi\epsilon}{\ln(b/a)} = 2\pi\epsilon$. (F/m).

P.3-22 $\bar{D} = \bar{a}_r \frac{\rho_\ell}{2\pi r}$. $\bar{E}_1 = \bar{a}_r \frac{\rho_\ell}{2\pi\epsilon_0\epsilon_{r1} r}$, $r_i < r < b$;

$\bar{E}_2 = \bar{a}_r \frac{\rho_\ell}{2\pi\epsilon_0\epsilon_{r2} r}$, $b < r < r_o$.

$$V = -\int_{r_o}^{r_i} \bar{E}\cdot d\bar{r} = \frac{\rho_\ell}{2\pi\epsilon_0}\left[\frac{1}{\epsilon_{r1}}\ln\left(\frac{b}{r_i}\right) + \frac{1}{\epsilon_{r2}}\ln\left(\frac{r_o}{b}\right)\right],$$

$$C' = \frac{\rho_\ell}{V} = \frac{2\pi\epsilon_0}{\frac{1}{\epsilon_{r1}}\ln\left(\frac{b}{r_i}\right) + \frac{1}{\epsilon_{r2}}\ln\left(\frac{r_o}{b}\right)} \quad \text{(F/m)}.$$

P.3-23 Assume charges $+Q$ and $-Q$ on the inner and outer conductors, respectively. $\bar{E} = \bar{a}_R E_R = \bar{a}_R \frac{Q}{4\pi\epsilon_0 R^2}$.

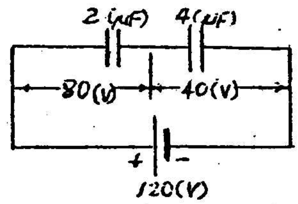
$$V = -\int_{R_o}^{R_i} \bar{E}\cdot \bar{a}_R\, dR = \frac{Q}{4\pi\epsilon}\left(\frac{1}{R_i} - \frac{1}{R_o}\right).$$

$$C = \frac{Q}{V} = \frac{4\pi\epsilon}{(1/R_i - 1/R_o)}.$$

P.3-24 Total capacitance across battery terminals,

$$C_T = \frac{4}{3}\ (\mu F).$$

Total stored elctric energy $W_T = \frac{1}{2} C_T (120)^2 = 9.6 (mW)$.



W_e in 2-(μF) capacitor $= \frac{1}{2}(2\times10^{-6})\times 80^2 = 6.4 (mW)$.

W_e in 1-(μF) capacitor $= \frac{1}{2}(10^{-6})\times 40^2 = 0.8 (mW)$.

W_e in 3-(μF) capacitor $= \frac{1}{2}(3\times10^{-6})\times 40^2 = 2.4 (mW)$.

P.3-25 $\bar{E} = \bar{a}_r 6r\sin\phi + \bar{a}_\phi 3r\cos\phi$,

$d\bar{\ell} = \bar{a}_r\, dr + \bar{a}_\phi r\, d\phi + \bar{a}_z dz$,

$$W_e = -Q\int_{P_1}^{P_2} \bar{E}\cdot d\bar{\ell} = -(5\times10^{-10})\left[6\sin\phi\int_2^4 r\, dr + 3r^2\int_{\pi/3}^{-\pi/2}\cos\phi\, d\phi\right].$$

a) First $r=2$, ϕ from $\pi/3$ to $-\pi/2$; then $\phi = -\pi/2$, r from 2 to 4:

$$W_e = -(5\times10^{-10})\left[3(2)^2\sin\phi\Big|_{\pi/3}^{-\pi/2} + 6\sin(-\tfrac{\pi}{2})\frac{r^2}{2}\Big|_2^4\right]$$
$$= -(5\times10^{-10})[-18-36] = 27\times10^{-9}(J) = 27 (nJ).$$

b) First $\phi = \pi/3$, r from 2 to 4; then $r=4$, ϕ from $\pi/3$ to $-\pi/2$:

$$W_e = -(5\times10^{-10})\left[6\sin(\tfrac{\pi}{3})\frac{r^2}{2}\Big|_2^4 + 3(4)^2\sin\phi\Big|_{\pi/3}^{-\pi/2}\right]$$
$$= -(5\times10^{-10})[18-72] = 27\times10^{-9}(J) = 27 (nJ).$$

—— Same as W_e in part a). $\bar{\nabla}\times\bar{E} = 0 \rightarrow \bar{E}$ is conservative.

P.3-26 Assume the inner and outer radii to be a and $a+r$ respectively. Substituting Eq.(3-89) in Eq.(3-117) and using Eq.(3-115), we have

$$F_Q = -\frac{\partial}{\partial r}\left(\frac{Q}{2}\cdot\frac{Q}{2\pi\epsilon L}\ln\frac{a+r}{a}\right)$$
$$= -\frac{Q^2}{4\pi\epsilon L(a+r)} = -\frac{Q^2}{4\pi\epsilon L b},$$ in the direction of decreasing r (attraction).

P.3-27 Switch open: Charges on the plates are constant.

$$Q = CV_0 \ , \quad W_e = \frac{Q^2}{2C}.$$
$$C = \frac{w}{d}\left[\epsilon x + \epsilon_0(L-x)\right].$$
$$\bar{F}_Q = -\bar{\nabla} W_e = -\bar{a}_x \frac{Q^2}{2}\frac{\partial}{\partial x}\left(\frac{1}{C}\right)$$
$$= \bar{a}_x \frac{Q^2 d}{2w}\,\frac{\epsilon - \epsilon_0}{[\epsilon x + \epsilon_0(L-x)]^2} = \bar{a}_x \frac{V_0^2 w}{2d}(\epsilon - \epsilon_0).$$

P.3-28 Use subscripts d and a to denote dielectric and air regions respectively. $\nabla^2 V = 0$ in both regions.

$$V_d = c_1 y + c_2, \quad \bar{E}_d = -\bar{a}_y c_1, \quad \bar{D}_d = -\bar{a}_y \epsilon_0\epsilon_r c_1.$$
$$V_a = c_3 y + c_4, \quad \bar{E}_a = -\bar{a}_y c_3, \quad \bar{D}_a = -\bar{a}_y \epsilon_0 c_3.$$

B.C.: At $y=0$, $V_d = 0$; at $y=d$, $V_a = V_0$;

at $y = 0.8d$: $V_d = V_a$, $\bar{D}_d = \bar{D}_a$.

Solving: $c_1 = \dfrac{V_0}{(0.8+0.2\epsilon_r)d}$, $c_2 = 0$, $c_3 = \dfrac{\epsilon_r V_0}{(0.8+0.2\epsilon_r)d}$, $c_4 = \dfrac{(1-\epsilon_r)V_0}{1+0.25\epsilon_r}$.

a) $V_d = \dfrac{5yV_0}{(4+\epsilon_r)d}$, $\quad \bar{E}_d = -\bar{a}_y \dfrac{5V_0}{(4+\epsilon_r)d}$.

b) $V_a = \dfrac{5\epsilon_r y - 4(\epsilon_r - 1)d}{(4+\epsilon_r)d} V_0$, $\quad \bar{E}_a = -\bar{a}_y \dfrac{5\epsilon_r V_0}{(4+\epsilon_r)d}$.

c) $(\rho_s)_{y=d} = -(D_a)_{y=d} = \dfrac{5\epsilon_0\epsilon_r V_0}{(4+\epsilon_r)d}$.

$(\rho_s)_{y=0} = (D_d)_{y=0} = -\dfrac{5\epsilon_0\epsilon_r V_0}{(4+\epsilon_r)d}$.

P.3-29 Poisson's eq. $\nabla^2 V = -\dfrac{A}{\epsilon r} \longrightarrow \dfrac{1}{r}\dfrac{\partial}{\partial r}\left(r\dfrac{\partial V}{\partial r}\right) = -\dfrac{A}{\epsilon r}$.

Solution: $V = -\dfrac{A}{\epsilon} r + c_1 \ln r + c_2$.

B.C.: $\begin{cases} \text{At } r=a, & V_0 = -\frac{A}{\epsilon}a + c_1 \ln a + c_2. \\ \text{At } r=b, & 0 = -\frac{A}{\epsilon}b + c_1 \ln b + c_2. \end{cases}$ $\quad c_1 = \dfrac{\frac{A}{\epsilon}(b-a) - V_0}{\ln(b/a)}$, $\quad c_2 = \dfrac{V_0 \ln b + \frac{A}{\epsilon}(a\ln b - b\ln a)}{\ln(b/a)}$.

P.3-30 $\bar{\nabla}^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0.$

Solution: $V = c_1 \ln r + c_2$.

Boundary conditions:

$$\text{At } r = a,\ V = V_0 = c_1 \ln a + c_2.$$

$$\text{At } r = 0,\ V = 0 = c_1 \ln b + c_2.$$

$$c_1 = -\frac{V_0}{\ln(b/a)}, \quad c_2 = \frac{V_0 \ln b}{\ln(b/a)}.$$

$$V = V_0 \frac{\ln(b/r)}{\ln(b/a)}, \quad \bar{E} = -\bar{\nabla} V = \bar{a}_r \frac{V_0}{r \ln(b/a)}.$$

Surface densities: At $r = a$, $\rho_{sa} = \epsilon_0 E_r = \frac{\epsilon_0 V_0}{a \ln(b/a)}$.

At $r = b$, $\rho_{sb} = -\epsilon_0 E_r = -\frac{\epsilon_0 V_0}{b \ln(b/a)}$.

Capacitance per unit length $C' = \frac{Q}{V_{ab}} = \frac{2\pi a \rho_{sa}}{V_0} = \frac{2\pi\epsilon_0}{\ln(b/a)}$ (C/m).

P.3-31 V and $\bar{E}$ depend only on θ. $\longrightarrow$ Eq.(3-129): $\frac{d}{d\theta}\left(\sin\theta\frac{dV}{d\theta}\right) = 0.$

a) Solution: $\frac{dV}{d\theta} = \frac{C_1}{\sin\theta} \longrightarrow V(\theta) = C_1 \ln\left(\tan\frac{\theta}{2}\right) + C_2.$

B.C. ① $V(\alpha) = V_0 = C_1 \ln\left(\tan\frac{\theta}{2}\right) + C_2.$

② $V\left(\frac{\pi}{2}\right) = 0 = C_1 \ln\left(\tan\frac{\pi}{4}\right) + C_2 \longrightarrow C_2 = 0.$

$\longrightarrow C_1 = \frac{V_0}{\ln[\tan(\alpha/2)]} \longrightarrow V(\theta) = \frac{V_0 \ln[\tan(\theta/2)]}{\ln[\tan(\alpha/2)]}.$

b) $\bar{E} = -\bar{a}_\theta \frac{dV}{R\,d\theta} = -\bar{a}_\theta \frac{V_0}{R \ln[\tan(\alpha/2)] \sin\theta}.$

c) On the cone: $\theta = \alpha$, $\rho_s = \epsilon_0 E(\alpha) = -\frac{\epsilon_0 V_0}{R \ln[\tan(\alpha/2)] \sin\alpha}.$

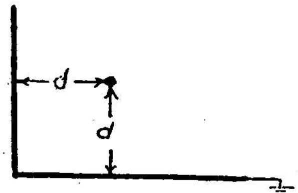
On the grounded plane: $\theta = \pi/2$, $\rho_s = -\epsilon_0 E\left(\frac{\pi}{2}\right) = \frac{\epsilon_0 V_0}{R \ln[\tan(\alpha/2)]}.$

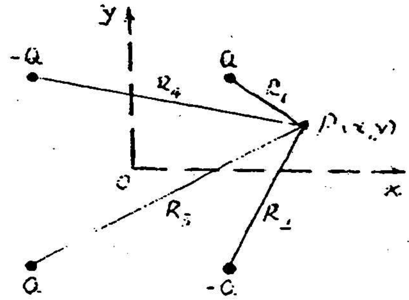
P.3-32 Consider the conditions in the xy-plane ($z = 0$).

a) $V_P = \frac{Q}{4\pi\epsilon}\left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3} - \frac{1}{R_4}\right)$, where

$$R_1 = [(x-d)^2 + (y-d)^2]^{1/2}, \quad R_2 = [(x-d)^2 + (y+d)^2]^{1/2},$$

$$R_3 = [(x+d)^2 + (y+d)^2]^{1/2}, \quad R_4 = [(x+d)^2 + (y-d)^2]^{1/2}.$$





$$\bar{E}_p = -\bar{\nabla} V_p = -\bar{a}_x \frac{\partial V_p}{\partial x} - \bar{a}_y \frac{\partial V_p}{\partial y}$$

$$= \bar{a}_x \frac{Q}{4\pi\epsilon}\left[-\frac{x-d}{R_1^3} + \frac{x-d}{R_2^3} - \frac{x+d}{R_3^3} + \frac{x+d}{R_4^3}\right]$$

$$+ \bar{a}_y \frac{Q}{4\pi\epsilon}\left[-\frac{y-d}{R_1^3} + \frac{y+d}{R_2^3} - \frac{y+d}{R_3^3} + \frac{y-d}{R_4^3}\right].$$

E_p will have a z-component if the point P does not lie in the xy-plane.

b) On the conducting half-planes, $\rho_s = D_n = \epsilon E_n$.
Along the x-axis, $y=0$: $R_1 = [(x-d)^2 + d^2]^{1/2} = R_2$,
and $R_3 = [(x+d)^2 + d^2]^{1/2} = R_4$:

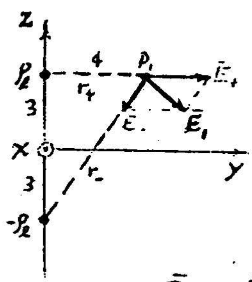
$$E_x = 0, \quad E_y = \frac{Q}{2\pi\epsilon}\left[\frac{d}{R_1^3} - \frac{d}{R_3^3}\right].$$

$$\therefore \rho_s(y=0) = \frac{Qd}{2\pi}\left\{\frac{1}{[(x-d)^2+d^2]^{3/2}} - \frac{1}{[(x+d)^2+d^2]^{3/2}}\right\}$$

$$= \begin{cases} 0, & \text{at } x=0. \\ \text{max.}, & \text{at } x=d. \end{cases}$$

Similarly for $\rho_s(x=0)$ on the vertical conducting plane by changing x to y.

P.3-34



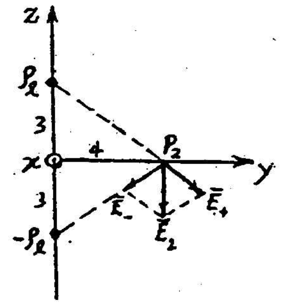
Assume ρ_ℓ (50 nC/m) to be at $y=0$ and $z=3$ (m)

a) Vector from ρ_ℓ to $P_1(0,4,3)$: $\bar{r}_+ = \bar{a}_y 4$.
Vector from $-\rho_\ell$ to P_1: $\bar{r}_- = \bar{a}_y 4 + \bar{a}_z 6$.

$$\bar{E}_+ = \frac{\rho_\ell}{2\pi\epsilon_0}\frac{\bar{r}_+}{r_+^2} = \frac{\rho_\ell}{2\pi\epsilon_0}\frac{\bar{a}_y}{4} = 9\times10^5(\bar{a}_y 0.25)$$

$$\bar{E}_- = \frac{-\rho_\ell}{2\pi\epsilon_0}\frac{\bar{r}_-}{r_-^2} = -9\times10^5(\bar{a}_y 0.077 + \bar{a}_z 0.115).$$

$$\bar{E}_1 = \bar{E}_+ + \bar{E}_- = 9\times10^5(\bar{a}_y 0.173 - \bar{a}_z 0.115) \quad \text{(V/m) at } P_1.$$



b) At P_2 (0,4,0) on the xy-plane (the ground):

Vector from ρ_ℓ to P_2 is $\bar{r}_+ = \bar{a}_y 4 - \bar{a}_z 3$,

Vector from $-\rho_\ell$ to P_2 is $\bar{r}_- = \bar{a}_y 4 + \bar{a}_z 3$.

$$\bar{E}_+ = \frac{\rho_\ell}{2\pi\epsilon_0}\,\frac{\bar{a}_y 4 - \bar{a}_z 3}{4^2+3^2}, \quad \bar{E}_- = \frac{-\rho_\ell}{2\pi\epsilon_0}\,\frac{\bar{a}_y 4 + \bar{a}_z 3}{4^2+3^2}.$$

$$\bar{E}_2 = \bar{E}_+ + \bar{E}_- = \frac{\rho_\ell}{2\pi\epsilon_0}\left(\frac{-\bar{a}_z 6}{4^2+3^2}\right)$$

$$= 9\times10^5(-\bar{a}_z 0.24) = -\bar{a}_z 2.16\times10^5 \text{ (V/m)}.$$

$$\rho_{s2} = \epsilon_0 E_{2z} = \frac{\rho_\ell}{2\pi}(-0.24) = \frac{50\times10^{-6}}{2\pi}(0.24) = -1.91\times10^{-6}\text{ (C/m}^2\text{)}$$

$$= -1.91\ (\mu\text{C/m}^2).$$

P.3-35 Given $D = 2$ (cm), $a = 0.3$ (cm).

a) From Eq. (3-163),

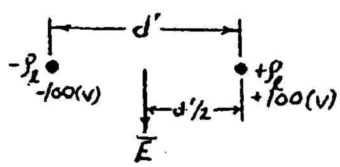
$$d = \frac{1}{2}\left(D + \sqrt{D^2 - 4a^2}\right) = \frac{1}{2}\left[2 + \sqrt{2^2 - 4(0.3)^2}\right] = 1.954 \text{ (cm)}.$$

$$d_i = D - d = 2 - 1.954 = 0.046 \text{ (cm)} = 0.46 \text{ (mm)}.$$

b) $$\rho_\ell = \frac{2\pi\epsilon_0 V_1}{\ln(d/a)} = \frac{2\pi\left(\frac{1}{36\pi}\times10^{-9}\right)\times100}{\ln(1.954/0.3)} = 2.96\times10^{-9}\text{ (F/m)}$$

$$= 2.96 \text{ (nF/m)}.$$

c) The equivalent line charges are separated by



$$d' = d - d_i = 1.954 - 0.046$$
$$= 1.908 \text{ (cm)}.$$

$$|\bar{E}| = \frac{\rho_\ell}{2\pi\epsilon_0(d'/2)}\times2 = 111.9 \text{ (V/m)},$$

— in a direction normal to the plane containing the wires.

Chapter 4

Steady Electric Currents

P.4-1 a) $R = \frac{\ell}{\sigma S} = \frac{V}{I} \longrightarrow \sigma = \frac{\ell I}{SV} = 3.54\times10^{7}\ (S/m).$

b) $E = \frac{V}{\ell} = 6\times10^{-3}\ (V/m).$

c) $P = VI = 1\ (W).$

d) $\rho_e = -\frac{\sigma}{\mu_e}$. The given electron mobility $1.4\times10^{-3}\ (m^2\cdot V/s)$ is that of a good conductor.

$$u = \left|\frac{J}{\rho_e}\right| = \left|\frac{\mu_e J}{\sigma}\right| = |\mu_e E| = 1.4\times10^{-3}\times(6\times10^{-3})$$
$$= 8.4\times10^{-6}\ (m/s).$$

P.4-2 R_1 = Resistance per unit length of core $= \frac{1}{\sigma S_1} = \frac{1}{\sigma\pi a^2}$.

R_2 = Resistance per unit length of coating $= \frac{1}{0.1\sigma S_2}$.

Let b = Thickness of coating. $\longrightarrow S_2 = \pi(a+b)^2 - \pi a^2 = \pi(2ab+b^2).$

a) $R_1 = R_2 \longrightarrow b = (\sqrt{11}-1)a = 2.32a.$

b) $I_1 = I_2 = \frac{I}{2}$. $\quad J_1 = \frac{I}{2\pi a^2}, \quad J_2 = \frac{I}{2S_2} = \frac{I}{20S_1} = \frac{I}{20\pi a^2}.$

$$E_1 = \frac{J_1}{\sigma} = \frac{I}{2\pi a^2\sigma}, \quad E_2 = \frac{J_2}{0.1\sigma} = \frac{I}{2\pi a^2\sigma}.$$

Thus, $J_1 = 10J_2$ and $E_1 = E_2$.

P.4-3 $\rho_0 = \frac{Q_0}{(4\pi/3)b^3} = \frac{10^{-3}}{(4\pi/3)(0.1)^3} = 0.239\ (C/m^3), \quad \rho = \rho_0 e^{-(\sigma/\epsilon)t}.$

a) $R < b:\ \bar{E}_i = \bar{a}_R \frac{(4\pi/3)R^3\rho}{4\pi\epsilon R^2} = \bar{a}_R \frac{\rho_0 R}{3\epsilon} e^{-(\sigma/\epsilon)t} = \bar{a}_R\, 7.5\times10^{9} R\, e^{-9.42\times10^{11}t}\ (V/m).$

$R > b:\ \bar{E}_o = \bar{a}_R \frac{Q_0}{4\pi\epsilon_0 R^2} = \bar{a}_R \frac{9}{R^2}\times10^{6}\ (V/m).$

b) $R < b:\ \bar{J}_i = \sigma\bar{E}_i = \bar{a}_R\, 7.5\times10^{10} R\, e^{-9.42\times10^{11}t}\ (A/m^2).$

$R > b:\ \bar{J}_o = 0.$

P.4-4 a) $e^{-(\sigma/\epsilon)t} = \frac{\rho}{\rho_0} = 0.01 \longrightarrow t = \frac{\ln 100}{(\sigma/\epsilon)} = 4.88\times10^{-12}\,(s) = 4.88\,(ps).$

b) $W_i = \frac{\epsilon}{2}\int_{V'} E_i^2\, dv' = \frac{2\pi\rho_0^2 b^5}{45\epsilon} e^{-2(\sigma/\epsilon)t} = (W_i)_0 \left[e^{-(\sigma/\epsilon)t}\right]^2.$

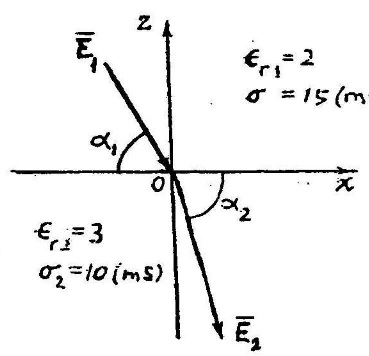
$\therefore \frac{W_i}{(W_i)_0} = \left[e^{-(\sigma/\epsilon)t}\right]^2 = 0.01^2 = 10^{-4}.$ Energy dissipated as heat loss.

c) Electrostatic energy stored outside the sphere $W_o = \frac{\epsilon_0}{2}\int_b^\infty E_o^2\, 4\pi R^2 dR = \frac{Q_0^2}{8\pi\epsilon_0 b} = 45\ (kJ)$ — Constant.

P.4-5 $I_1 = 0.1\,(A)$, $P_{R1} = 3.33\,(mW)$; $I_2 = 0.02\,(A)$, $P_{R2} = 8.00\,(mW)$;
$I_3 = 0.0133\,(A)$, $P_{R3} = 5.31\,(mW)$; $I_4 = 0.0333\,(A)$, $P_{R4} = 8.87\,(mW)$;
$I_5 = 0.0667\,(A)$, $P_{R5} = 44.5\,(mW)$. $\sum_n P_{Rn} = V_0 I_1 = 70\,(mW).$

Total resistance seen by the source $= 7\ (\Omega)$.

P.4-6



$\bar{E}_1 = \bar{a}_x 20 - \bar{a}_z 50\ \ (V/m).$

a) $E_{2t} = E_{1t} = 20.$

$J_{2n} = J_{1n} \longrightarrow \sigma_2 E_{2n} = \sigma_1 E_{1n}$

$\longrightarrow E_{2n} = \frac{\sigma_1}{\sigma_2} E_{1n} = \frac{15}{10}(-50) = -75.$

$\therefore \bar{E}_2 = \bar{a}_x 20 - \bar{a}_z 75\ \ (V/m).$

b) $\bar{J}_1 = \sigma_1 \bar{E}_1 = 15\times10^{-3}(\bar{a}_x 20 - \bar{a}_z 50) = \bar{a}_x 0.3 - \bar{a}_z 0.75\ \ (A/m^2).$

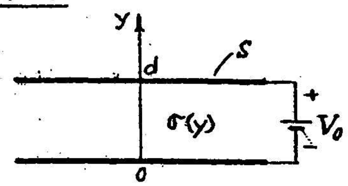
$\bar{J}_2 = \sigma_2 \bar{E}_2 = 10\times10^{-3}(\bar{a}_x 20 - \bar{a}_z 75) = \bar{a}_x 0.2 - \bar{a}_z 0.75\ \ (A/m^2).$

c) $\alpha_1 = \tan^{-1}\left(\frac{50}{20}\right) = 68.2^\circ,\qquad \alpha_2 = \tan^{-1}\left(\frac{75}{20}\right) = 75.1^\circ.$

d) $D_{2n} - D_{1n} = \rho_s \longrightarrow \epsilon_2 E_{2n} - \epsilon_1 E_{1n} = \rho_s.$

$\rho_s = \epsilon_0(-3\times75 + 2\times50) = -125\epsilon_0 = -1.105\ (nC/m^2).$

P.4-7



$$\sigma(y) = \sigma_1 + (\sigma_2 - \sigma_1)\frac{y}{d}.$$

a) Neglecting fringing effect and assuming a current density:

$$\bar{J} = -\bar{a}_y J_0 \longrightarrow \bar{E} = \frac{\bar{J}}{\sigma} = -\bar{a}_y \frac{J_0}{\sigma(y)}.$$

$$V_0 = -\int_0^d \bar{E}\cdot\bar{a}_y\,dy = \int_0^d \frac{J_0\,dy}{\sigma_1 + (\sigma_2-\sigma_1)\frac{y}{d}} = \frac{J_0 d}{\sigma_2-\sigma_1}\ln\frac{\sigma_2}{\sigma_1}.$$

$$R = \frac{V_0}{I} = \frac{V_0}{J_0 S} = \frac{d}{(\sigma_2-\sigma_1)S}\ln\frac{\sigma_2}{\sigma_1}.$$

b) $(\rho_s)_u = \epsilon_0 E_y(d) = \frac{\epsilon_0 J_0}{\sigma_2} = \frac{\epsilon_0(\sigma_2-\sigma_1)V_0}{\sigma_2 d\,\ln(\sigma_2/\sigma_1)}$ on upper plate,

$(\rho_s)_\ell = -\epsilon_0 E_y(0) = -\frac{\epsilon_0 J_0}{\sigma_1} = -\frac{\epsilon_0(\sigma_2-\sigma_1)V_0}{\sigma_1 d\,\ln(\sigma_2/\sigma_1)}$ on lower plate.

P.4-8 a) Continuity of the normal component of $\bar{J}$ assures the same current in both media. By Kirchhoff's voltage law:

$$V_0 = (R_1 + R_2)I = \left(\frac{d_1}{\sigma_1 S} + \frac{d_2}{\sigma_2 S}\right)I.$$

$$\therefore\quad J = \frac{I}{S} = \frac{V_0}{(d_1/\sigma_1)+(d_2/\sigma_2)} = \frac{\sigma_1\sigma_2 V_0}{\sigma_2 d_1 + \sigma_1 d_2}.$$

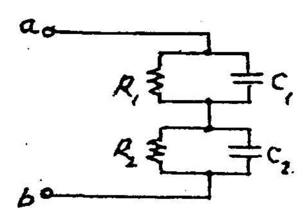
b) Two equations are needed for the determination of $\bar{E}_1$ and $\bar{E}_2$: $\quad V_0 = E_1 d_1 + E_2 d_2$

and $\quad \sigma_1 E_1 = \sigma_2 E_2.$

Solving, we have $E_1 = \frac{\sigma_2 V_0}{\sigma_2 d_1 + \sigma_1 d_2}$

and $E_2 = \frac{\sigma_1 V_0}{\sigma_2 d_1 + \sigma_1 d_2}.$

c) Equivalent R-C circuit between terminals a and b:



$$R_1 = \frac{d_1}{\sigma_1 S}.\quad R_2 = \frac{d_2}{\sigma_2 S}.\quad C_1 = \frac{\epsilon_1 S}{d_1}.\quad C_2 = \frac{\epsilon_2 S}{d_2}.$$

P. 4-9 a) Same equivalent R-C circuit as that in Problem P.4-8 with

$$R_1 = \frac{1}{2\pi\sigma_1 L}\ln\left(\frac{c}{a}\right), \qquad R_2 = \frac{1}{2\pi\sigma_2 L}\ln\left(\frac{b}{c}\right).$$

$$C_1 = \frac{2\pi\epsilon_1 L}{\ln(c/a)}, \qquad C_2 = \frac{2\pi\epsilon_2 L}{\ln(b/c)}.$$

b) $$I = V_0 G = V_0 \frac{1}{R_1+R_2} = \frac{2\pi\sigma_1\sigma_2 L V_0}{\sigma_1 \ln(b/c)+\sigma_2\ln(c/a)}.$$

$$J_1 = J_2 = \frac{I}{2\pi r L} = \frac{\sigma_1\sigma_2 V_0}{r\,[\sigma_1 \ln(b/c)+\sigma_2\ln(c/a)]}.$$

P. 4-10 Resistance $R = \frac{\ell}{\sigma S}$. (Eq. 4-16)

— Homogeneous material with a uniform cross section.

Between top and bottom flat faces: $S = \frac{\pi}{4}(b^2-a^2)$.

$$R = \frac{4h}{\sigma\pi(b^2-a^2)}.$$

P. 4-11 Use Laplace's equation in cylindrical coordinates.

$$\nabla^2 V = 0 \longrightarrow \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0.$$

Solution: $V(r) = c_1 \ln r + c_2$.

Boundary conditions: $V(a) = V_0$; $\qquad V(b) = 0$.

$$\longrightarrow V(r) = V_0 \frac{\ln(b/r)}{\ln(b/a)}.$$

$$\bar{E}(r) = -\bar{a}_r \frac{\partial V}{\partial r} = \bar{a}_r \frac{V_0}{r\ln(b/a)}.$$

$$\bar{J}(r) = \sigma \bar{E}(r).$$

$$I = \int_S \bar{J}\cdot d\bar{s} = \int_0^{\pi/2} \bar{J}\cdot(\bar{a}_r h r\, d\phi) = \frac{\pi\sigma h V_0}{2\ln(b/a)}.$$

$$R = \frac{V_0}{I} = \frac{2\ln(b/a)}{\pi\sigma h}.$$

P. 4-12 Assume a potential difference V_0 between the inner and outer spheres.

$$\bar{\nabla}^2 V = 0 \rightarrow \frac{1}{R^2}\frac{d}{dR}(R^2 V) = 0. \longrightarrow V = \frac{K}{R} \longrightarrow E_R = \frac{K}{R^2}.$$

$$V_0 = -\int_{R_2}^{R_1} E_R \, dR = -K\int_{R_2}^{R_1} \frac{1}{R^2} dR = K\left(\frac{1}{R_1} - \frac{1}{R_2}\right).$$

$$\longrightarrow K = \frac{V_0}{\frac{1}{R_1} - \frac{1}{R_2}}. \qquad J_R = \sigma E_R = \frac{\sigma V_0}{\frac{1}{R_1} - \frac{1}{R_2}}.$$

$$I = \int_0^{2\pi}\int_0^{\pi} J_R R^2 \sin\theta \, d\theta \, d\phi = \frac{4\pi\sigma V_0}{\frac{1}{R_1} - \frac{1}{R_2}}.$$

$$R = \frac{V_0}{I} = \frac{1}{4\pi\sigma}\left(\frac{1}{R_1} - \frac{1}{R_2}\right).$$

Chapter 5

Static Magnetic Fields

P.5-1 $\bar{F} = Q(\bar{E} + \bar{u}\times\bar{B}) = 0.$

$$\bar{E} = -\bar{u}\times\bar{B} = -\bar{a}_x u_0 \times (\bar{a}_x B_x + \bar{a}_y B_y + \bar{a}_z B_z)$$
$$= u_0(\bar{a}_y B_z - \bar{a}_z B_y).$$

P.5-2 $\bar{B} = \bar{a}_\phi B_\phi = \bar{a}_\phi \dfrac{\mu_0 N I}{2\pi r}.$

$$\Phi = \int_s B_\phi\, ds = \frac{\mu_0 N I}{2\pi}\int_a^b \frac{h}{r}\, dr$$
$$= \frac{\mu_0 N I h}{2\pi}\ln\frac{b}{a}.$$

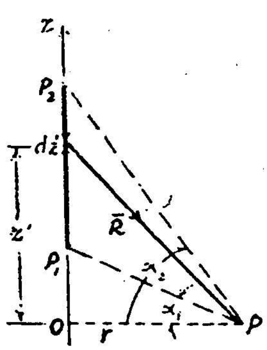
If B_ϕ at $r = \dfrac{a+b}{2}$ is used, $\Phi' = \dfrac{\mu_0 N I h}{\pi}\left(\dfrac{b-a}{b+a}\right).$

$$\%\ \text{error} = \frac{\Phi' - \Phi}{\Phi}\times 100$$
$$= \left[\frac{2(b-a)}{(b+a)\ln(b/a)} - 1\right]\times 100.$$

For $\dfrac{b}{a} = 5$, the error is $\left[\dfrac{2(5-1)}{(5+1)\ln 5} - 1\right]\times 100,$

or $-17.2\,\%$ (too low).

P.5-3 a) Use Eq. (5-32c). $d\bar{\ell}' = \bar{a}_z dz'$, $\bar{R} = \bar{a}_r r - \bar{a}_z z'$.



$$d\bar{\ell}'\times\bar{R} = \bar{a}_z dz' \times(\bar{a}_r r - \bar{a}_z z') = \bar{a}_\phi r\, dz'.$$
$$\bar{B}_P = \bar{a}_\phi \frac{\mu_0 I}{4\pi}\int \frac{r\, dz'}{(z'^2 + r^2)^{3/2}}.$$

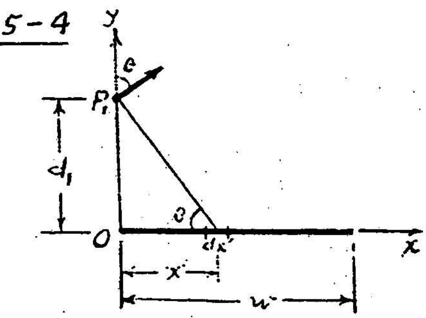
Let $z' = r\tan\alpha$, $dz' = r\sec^2\alpha\, d\alpha$.

$$\bar{B}_P = \bar{a}_\phi \frac{\mu_0 I}{4\pi r}\int_{\alpha_1}^{\alpha_2}\cos\alpha\, d\alpha$$
$$= \bar{a}_\phi \frac{\mu_0 I}{4\pi r}(\sin\alpha_2 - \sin\alpha_1).$$

b) For an infinitely long wire: $\alpha_2 \to 90°$ and $\alpha_1 \to -90°$.

$\bar{B}_P$ becomes $\bar{a}_\phi \dfrac{\mu_0 I}{2\pi r}$, as in Eq. (5-35).

P. 5-4



Use Eq. (5-35):

$$d\bar{B}_{P_1} = \bar{a}_x dB_x + \bar{a}_y dB_y$$
$$= \bar{a}_x (dB_{P_1}) \sin\theta + \bar{a}_y (dB_{P_1}) \cos\theta,$$

Where $dB_{P_1} = \dfrac{\mu_0 (I/w)\, dx'}{2\pi (x'^2 + d_1^2)^{1/2}}$,

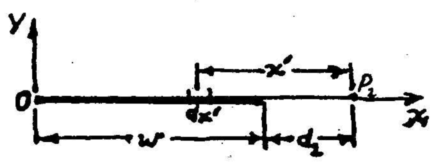
$$\sin\theta = \frac{d}{(x'^2+d^2)^{1/2}}, \quad \cos\theta = \frac{x'}{(x'^2+d^2)^{1/2}}.$$

$$\bar{B}_{P_1} = \bar{a}_x B_x + \bar{a}_y B_y,$$

where $\quad B_x = \dfrac{\mu_0 I d}{2\pi w} \displaystyle\int_0^w \frac{dx'}{x'^2+d^2} = \frac{\mu_0 I}{2\pi w} \tan^{-1}\left(\frac{w}{d}\right),$

and $\quad B_y = \dfrac{\mu_0 I}{2\pi w} \displaystyle\int_0^w \frac{x'\,dx'}{x'^2+d^2} = \frac{\mu_0 I}{4\pi w} \ln\left(1+\frac{w}{d}\right).$

P. 5-5

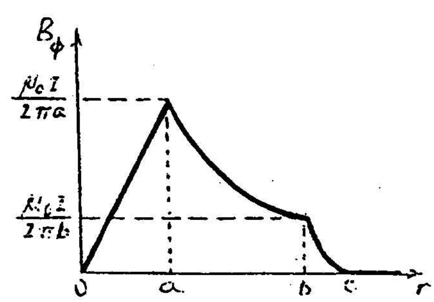


I flows into the paper (in $-\bar{a}_z$ direction).

$$d\bar{B}_{P_2} = -\bar{a}_z \frac{\mu_0 I\, dx'}{2\pi w x'}.$$

$$\bar{B}_{P_2} = -\bar{a}_z \frac{\mu_0 I}{2\pi w} \int_{d_2}^{d_2+w} \frac{dx'}{x'} = -\bar{a}_z \frac{\mu_0 I}{2\pi w} \ln\left(1+\frac{w}{d_2}\right).$$

P. 5-6 Apply Ampère's circuital law, Eq. (5-10), and assume the medium to be nonmagnetic:



$$\oint \bar{B} \cdot d\bar{\ell} = \mu_0 I.$$

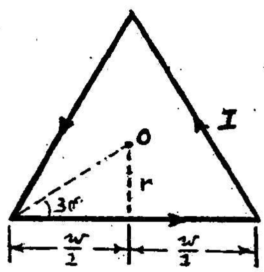
For $0 \le r \le a$, $\bar{B} = \bar{a}_\phi \dfrac{\mu_0 r I}{2\pi a^2}$.

For $a \le r \le b$, $\bar{B} = \bar{a}_\phi \dfrac{\mu_0 I}{2\pi r}$.

For $b \le r \le c$, $\bar{B} = \bar{a}_\phi \left(\dfrac{c^2 - r^2}{c^2 - b^2}\right) \dfrac{\mu_0 I}{2\pi r}$.

For $r \ge c$, $\bar{B} = 0$.

P. 5-7



Assume that the current flows in the counterclockwise direction in a triangle lying in the xy-plane. From Eq. (5-34) and noting that

$$L = \frac{w}{2} \quad \text{and} \quad r = \frac{w}{2}\tan 30° = \frac{w}{2\sqrt{3}},$$

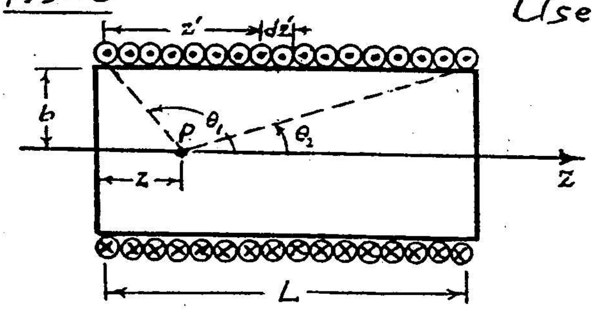
we have

$$\bar{B} = 3\left(\bar{a}_z \frac{\mu_0 I L}{2\pi r\sqrt{L^2+r^2}}\right) \text{ at } O.$$

$$L/r = \sqrt{3}, \quad \sqrt{L^2+r^2} = \frac{w}{\sqrt{3}}.$$

$$\bar{B} = \bar{a}_z \frac{3\mu_0 I}{2\pi}\,\frac{\sqrt{3}}{w/\sqrt{3}} = \bar{a}_z \frac{9\mu_0 I}{2\pi w}.$$

P. 5-8



Use Eq. (5-37):

$$d\bar{B} = \bar{a}_z \frac{\mu I b^2}{2[(z'-z)^2+b^2]^{3/2}}\left(\frac{N}{L}\right)dz',$$

$$\bar{B} = \bar{a}_z \frac{\mu N I b^2}{2L}$$

$$= \bar{a}_z \frac{\mu N I}{2L}\left[\frac{L-z}{\sqrt{(L-z)^2+b^2}} - \frac{z}{\sqrt{z^2+b^2}}\right]$$

$$= \bar{a}_z \frac{\mu N I}{2L}(\cos\theta_2 - \cos\theta_1).$$

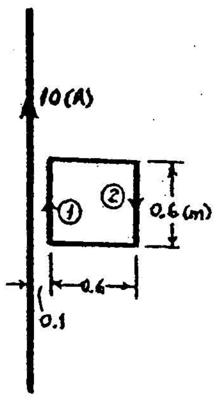
For $L \to \infty$, $\theta_2 \to 0$ and $\theta_1 \to \pi$, and $\bar{B} \to \bar{a}_z \frac{\mu N I}{L} = \bar{a}_z \mu n I$.

P. 5-9 a) $\bar{B} = \bar{\nabla}\times\bar{A} = \bar{a}_\phi \frac{\mu_0 I}{2\pi r} = -\bar{a}_\phi \frac{\partial A_z}{\partial r}$. (No change with z.)

$$\frac{dA_z}{dr} = -\frac{\mu_0 I}{2\pi r} \longrightarrow A_z = -\frac{\mu_0 I}{2\pi}\ln r + c.$$

$$A_z = 0 \text{ at } r = r_0. \longrightarrow c = \frac{\mu_0 I}{2\pi}\ln r_0.$$

$$\therefore \quad \bar{A} = \bar{a}_z A_z = \bar{a}_z \frac{\mu_0 I}{2\pi}\ln\left(\frac{r_0}{r}\right).$$



b) Use $\Phi = \oint \bar{A} \cdot d\bar{\ell}$.

Horizontal sides have no effect.

Side ①: $\int \bar{A} \cdot d\bar{\ell} = \left(\frac{\mu_0 I}{2\pi} \ln \frac{r_0}{0.1}\right) \times 0.6$.

Side ③: $\int \bar{A} \cdot d\bar{\ell} = -\left(\frac{\mu_0 I}{2\pi} \ln \frac{r_0}{0.7}\right) \times 0.6$.

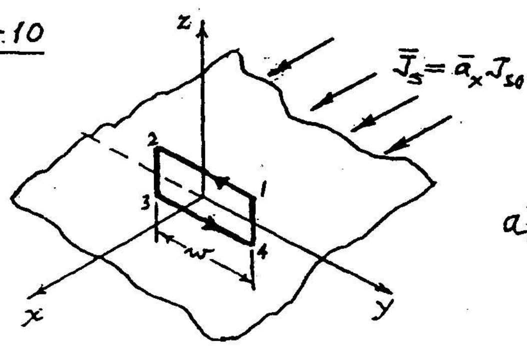
$$\oint \bar{A} \cdot d\bar{\ell} = \left(\frac{\mu_0 I}{2\pi} \ln \frac{0.7}{0.1}\right) \times 0.6$$

$$= \frac{(4\pi 10^{-7}) \times 10 \times 0.6}{2\pi} \ln 7$$

$$= 2.34 \times 10^{-6} \text{ (Wb)}.$$

$$\therefore \ \Phi = 2.34 \ (\mu\text{Wb}).$$

P.5-10



Infinite current sheet $\longrightarrow \bar{B}$ antisymmetrical and independent of x and y.

a) Apply Ampère's circuital law to path 12341:

$$\oint_C \bar{B} \cdot d\bar{\ell} = \mu_0 I \longrightarrow 2wB_y = \mu_0 J_{s0} w.$$

$$\longrightarrow B_y = \begin{cases} -\mu_0 J_{s0}/2 & \text{at } (0,0,z), \\ +\mu_0 J_{s0}/2 & \text{at } (0,0,-z). \end{cases}$$

or, $\quad \bar{B} = \frac{\mu_0}{2} \bar{J}_s \times \bar{a}_n$.

b) For $z > 0$, $\quad \bar{\nabla} \times \bar{A} = \bar{B} = \bar{a}_y\left(-\frac{\mu_0 J_{s0}}{2}\right)$.

$\bar{A}$ is independent of x and y.

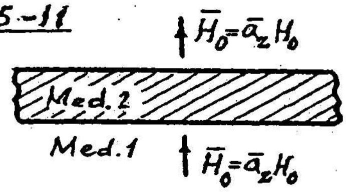
$$\frac{dA_x}{dz} = -\frac{\mu_0 J_{s0}}{2}.$$

$$A_x = -\frac{\mu_0 J_{s0}}{2} z + c.$$

At $z = z_0$, $A_x = 0 = -\frac{\mu_0 J_{s0}}{2} z_0 + c \longrightarrow c = \frac{\mu_0 J_{s0}}{2} z_0$.

$$\therefore \ \bar{A} = -\frac{\mu_0}{2}(z - z_0)\bar{J}_s.$$

P. 5-11



a) Given $\bar{B}_2=\mu_2\bar{H}_2$.

$$B_{2z}=B_{1z} \longrightarrow \mu H_2=\mu_0 H_0 \longrightarrow \bar{H}_2=\bar{a}_z H_2=\bar{a}_z\frac{\mu_0}{\mu}H_0.$$

b) Given $\bar{B}_2=\mu_0(\bar{H}_2+\bar{M}_i)$.

$$B_{2z}=B_{1z} \longrightarrow \mu_0(H_2+M_i)=\mu_0 H_0 \longrightarrow \bar{H}_2=\bar{a}_z(H_0-M_i).$$

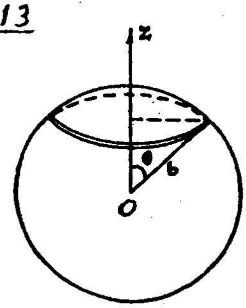
P. 5-12 a) $r<a$: $\left.\begin{array}{l}\bar{H}=\bar{a}_z nI,\\ \bar{B}=\bar{a}_z\mu nI\end{array}\right\}$ Eq. (6-14).

$$\bar{M}=\frac{\bar{B}}{\mu_0}-\bar{H}=\bar{a}_z\left(\frac{\mu}{\mu_0}-1\right)nI.$$

$a<r<b$: $\bar{H}=\bar{a}_z nI$, $\bar{B}=\bar{a}_z\mu_0 nI$, $\bar{M}=0$.

b) $\bar{J}_m=\bar{\nabla}\times\bar{M}=0$; $\quad \bar{J}_{ms}=\bar{M}\times\bar{a}_n=(\bar{a}_z\times\bar{a}_r)\left(\frac{\mu}{\mu_0}-1\right)nI=\bar{a}_\phi\left(\frac{\mu}{\mu_0}-1\right)nI.$

P. 5-13



$\bar{M}=\bar{a}_z M_0$

a) $\bar{J}_m=\bar{\nabla}\times\bar{M}=0.$

$$\bar{J}_{ms}=(\bar{a}_R\cos\theta-\bar{a}_\theta\sin\theta)M\times\bar{a}_R$$
$$=\bar{a}_\phi M_0\sin\theta.$$

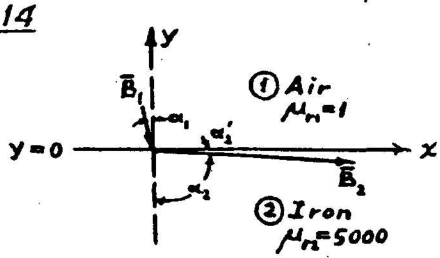
b) Apply Eq. (5-37) to a loop of radius $b\sin\theta$ carrying a current $J_{ms}\,b\,d\theta$:

$$d\bar{B}=\bar{a}_z\frac{\mu_0(J_{ms}b\,d\theta)(b\sin\theta)^2}{2(b^2)^{3/2}}$$
$$=\bar{a}_z\frac{\mu_0 M_0}{2}\sin^3\theta.$$

$$\bar{B}=\int d\bar{B}=\bar{a}_z\frac{\mu_0 M_0}{2}\int_0^\pi \sin^3\theta\,d\theta=\bar{a}_z\frac{2}{3}\mu_0 M_0=\frac{2}{3}\mu_0\bar{M},$$

at the center O.

P. 5-14



a) $\bar{B}_1=\bar{a}_x 2-\bar{a}_y 10$ (mT),

$\bar{B}_2=\bar{a}_x B_{2x}-\bar{a}_y B_{2y}$.

$$H_{2x}=\frac{B_{2x}}{5000\mu_0}=H_{1x}=\frac{2}{\mu_0}$$
$$\longrightarrow B_{2x}=10{,}000\ (\text{mT}),$$

$B_{2y}=B_{1y}=-10$ (mT).

$\therefore\ \bar{B}_2=\bar{a}_x 10{,}000-\bar{a}_y 10$ (mT).

$$\tan\alpha_2=\frac{\mu_2}{\mu_1}\tan\alpha_1=5000\left(\frac{B_{1x}}{B_{1y}}\right)=1{,}000 \longrightarrow \alpha_2=89.94°,\ \alpha_2'=0.04°.$$

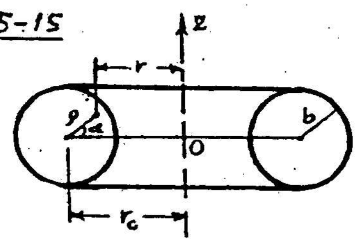
b) If $\bar{B}_2 = \bar{a}_x 10 + \bar{a}_y$ (mT), $\bar{B}_1 = \bar{a}_x B_{1x} + \bar{a}_y B_{1y}$.

$$H_{1x} = \frac{B_{1x}}{\mu_1} = H_{2x} = \frac{B_{2x}}{\mu_2} \longrightarrow B_{1x} = \frac{1}{\mu_{r2}} B_{2x} = \frac{10}{5000} = 0.002.$$

$$B_{1y} = B_{2y} = 2. \qquad \therefore \bar{B}_1 = \bar{a}_x 0.002 + \bar{a}_y 2 \quad (\text{mT}).$$

$$\alpha_1 = \tan^{-1}\frac{B_{1x}}{B_{1y}} \cong \frac{0.002}{2} = 0.001 \ (\text{rad}) = 0.057°$$

P. 5-15



$$\bar{B} = \bar{a}_\phi B_\phi = \bar{a}_\phi \frac{\mu_0 NI}{2\pi r}, \quad r = r_o - \rho\cos\alpha.$$

$$\Phi = \frac{\mu_0 NI}{2\pi}\int_0^b\int_0^{2\pi} \frac{\rho\, d\alpha\, d\rho}{r_o - \rho\cos\alpha} = \mu_0 NI\left(r_o - \sqrt{r_o^2 - b^2}\right).$$

$$\therefore L = \frac{N\Phi}{I} = \mu_0 N^2\left(r_o - \sqrt{r_o^2 - b^2}\right).$$

If $r_o \gg b$, $B_\phi \cong \frac{\mu_0 NI}{2\pi r_o}$ (constant).

$$\Phi \cong B_\phi S = B_\phi(\pi b^2) = \frac{\mu_0 N b^2 I}{2 r_o} \longrightarrow L \cong \frac{\mu_0 N^2 b^2}{2 r_o}.$$

P. 5-16 For I in the long straight wire, $\bar{B} = \bar{a}_\phi \frac{\mu_0 I}{2\pi r}$.

$$\Lambda_{12} = \int_S \bar{B}\cdot d\bar{s} = \int B_\phi \frac{2}{\sqrt{3}}(r-d)\,dr = \frac{\mu_0 I}{\pi\sqrt{3}}\int_d^{d+\frac{\sqrt{3}}{2}b}\left(\frac{r-d}{r}\right)dr$$

$$= \frac{\mu_0 I}{\pi\sqrt{3}}\left[\frac{\sqrt{3}}{2}b - d\ln\left(1 + \frac{\sqrt{3}b}{2d}\right)\right],$$

$$L_{12} = \frac{\Lambda_{12}}{I} = \frac{\mu_0}{\pi}\left[\frac{b}{2} - \frac{d}{\sqrt{3}}\ln\left(1+\frac{\sqrt{3}b}{2d}\right)\right].$$

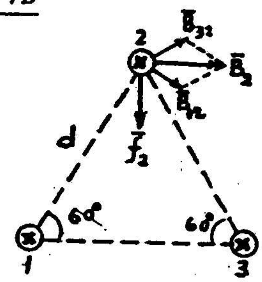
P. 5-17 Approximate the magnetic flux due to the long loop linking with the small loop by that due to two infinitely long wires carrying equal and opposite current I.

$$\Lambda_{12} = \frac{\mu_0 h_2 I}{2\pi}\int_0^{w_2}\left(\frac{1}{d+x} - \frac{1}{w_1+d+x}\right)dx$$

$$= \frac{\mu_0 h_2 I}{2\pi}\ln\left(\frac{w_2+d}{d}\cdot\frac{w_1+d}{w_1+w_2+d}\right).$$

$$L_{12} = \frac{\Lambda_{12}}{I} = \frac{\mu_0 h_2}{2\pi}\ln\frac{(w_1+d)(w_2+d)}{d(w_1+w_2+d)}.$$

P. 5-18



$I_1 = I_2 = I_3 = 25$ (A) ; $d = 0.15$ (m).

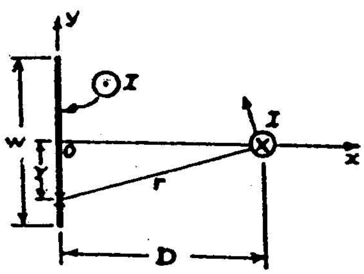
$\bar{B}_2 = \bar{a}_x 2B_{12}\cos 30° = \bar{a}_x \frac{\sqrt{3}\mu_0 I}{2\pi d}$.

Force per unit length on wire 2:

$$\bar{f}_2 = -\bar{a}_y I B_2 = -\bar{a}_y \frac{\sqrt{3}\mu_0 I^2}{2\pi d}$$
$$= -\bar{a}_y 1150\mu_0 = -\bar{a}_y 1.44\times 10^{-3}\ \text{(N/m)}.$$

Forces on all three wires are of equal magnitude and toward the center of the triangle.

P. 5-19 Magnetic field intensity at the wire due to the current $dI = \frac{I}{w}dy$ in an elemental dy is



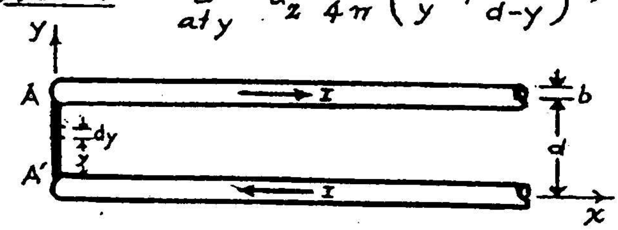
$$|d\bar{H}| = \frac{dI}{2\pi r} = \frac{I\,dy}{2\pi w\sqrt{D^2+y^2}}.$$

Symmetry $\longrightarrow$ $\bar{H}$ at the wire has only a y-component.

$$\bar{H} = \bar{a}_y \int (dH)\cdot\left(\frac{D}{r}\right) = \bar{a}_y 2\int_0^{w/2} \frac{ID\,dy}{2\pi w(D^2+y^2)}$$
$$= \bar{a}_y \frac{I}{\pi w}\tan^{-1}\left(\frac{w}{2D}\right).$$

$$\bar{f}' = \bar{I}\times\bar{B} = (-\bar{a}_z I)\times(\mu_0\bar{H}) = \bar{a}_x \frac{\mu_0 I^2}{\pi w}\tan^{-1}\left(\frac{w}{2D}\right)\ \text{(N/m)}.$$

P. 5-20 $\underset{\text{at } y}{\bar{B}} = -\bar{a}_z \frac{\mu_0 I}{4\pi}\left(\frac{1}{y} + \frac{1}{d-y}\right)$, $d\bar{\ell} = \bar{a}_y dy$.

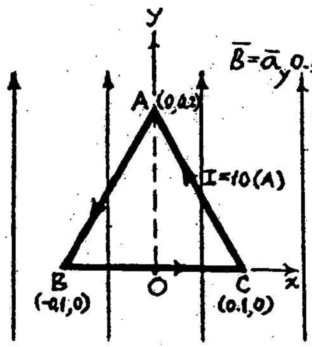


(A rail-gun problem.)

$$d\bar{F} = I d\bar{\ell}\times\bar{B}$$
$$= -\bar{a}_x \frac{\mu_0 I^2}{4\pi}\left(\frac{1}{y} + \frac{1}{d-y}\right)dy$$
$$\bar{F} = -\bar{a}_x \frac{\mu_0 I^2}{4\pi}\int_b^{d-b}\left(\frac{1}{y} + \frac{1}{d-y}\right)dy$$
$$= -\bar{a}_x \frac{\mu_0 I^2}{2\pi}\ln\left(\frac{d}{b} - 1\right).$$

P. 5-21 Force in a uniform magnetic field:

$$\bar{F} = I\bar{L} \times \bar{B} = -\bar{B} \times (I\bar{L}).$$



$\bar{B} = \bar{a}_y 0.5\,(T)$. $\quad -\bar{B}\times \quad I(\overrightarrow{AB}) \qquad I(\overrightarrow{CA}) \qquad I(\overrightarrow{BC})$

$-(\bar{a}_y 0.5) \times 10(-\bar{a}_x 0.1 - \bar{a}_y 0.2), \quad 10(-\bar{a}_x 0.1 + \bar{a}_y 0.2), \quad 10\bar{a}_x 0.2$

<u>Force:</u> $-\bar{a}_z 0.5 \qquad -\bar{a}_z 0.5 \qquad \bar{a}_z 1.0 \;(N)$

<u>Torque on loop</u>:

$$\bar{T} = \bar{m} \times \bar{B} = (\bar{a}_z IS) \times \bar{B}$$
$$= (\bar{a}_z 10 \times \tfrac{1}{2} \times 0.2 \times 0.2) \times (\bar{a}_y 0.5) = -\bar{a}_x 0.1\;(N\cdot m).$$

P. 5-22 $\bar{B}_2$ at the center of the large circular turn of wire carrying a current I_2 is (by setting $z = 0$ in Eq. 5-37):

$$\bar{B}_2 = \bar{a}_{n2} \frac{\mu_0 I_2}{2 r_2}.$$

Torque on the small circular turn of wire carrying a current I_1 is

$$\bar{T} = \bar{m}_1 \times \bar{B}_2 \cong (\bar{a}_{n1} I_1 \pi r_1^2) \times (\bar{a}_{n2} \frac{\mu_0 I_2}{2 r_2})$$
$$= (\bar{a}_{n1} \times \bar{a}_{n2}) \frac{\mu_0 I_1 I_2 \pi r_1^2}{2 r_2},$$

which is a torque having a magnitude $\frac{\mu_0 I_1 I_2 \pi r_1^2}{2 r_2} \sin\theta$ and a direction tending to align the magnetic fluxes produced by I_1 and I_2.

Chapter 6

Time-Varying Fields and Maxwell's Equations

P.6-1

$$\mathcal{V} = -\int_S \frac{\partial \bar{B}}{\partial t}\cdot d\bar{s}$$
$$= -\int_S \frac{\partial}{\partial t}(\bar{\nabla}\times\bar{A})\cdot d\bar{s}$$
$$= -\oint \frac{\partial \bar{A}}{\partial t}\cdot d\bar{\ell}.$$

P.6-2

$$\bar{B} = \bar{a}_z\, 3\cos(5\pi 10^7 t - \tfrac{1}{3}\pi y)\times 10^{-6}\ \ (T).$$
$$\int_S \bar{B}\cdot d\bar{s} = \int_0^{0.3} \bar{a}_z\, 3\cos(5\pi 10^7 t - \tfrac{1}{3}\pi y)10^{-6}\cdot(\bar{a}_z 0.1 dy)$$
$$= -\frac{0.9}{\pi}\left[\sin(5\pi 10^7 t - 0.1\pi) - \sin 5\pi 10^7 t\right]\times 10^{-6}\ (Wb).$$
$$\mathcal{V} = -\frac{d}{dt}\int_S \bar{B}\cdot d\bar{s} = 4.5\left[\cos(5\pi 10^7 t - 0.1\pi) - \cos 5\pi 10^7 t\right]\ (V).$$
$$i = \frac{\mathcal{V}}{2R} = 0.15\left[\cos(5\pi 10^7 t - 0.1\pi) - \cos 5\pi 10^7 t\right]$$
$$= 0.023\sin(5\pi 10^7 t - 0.05\pi)\quad (A)$$
$$= 23\sin(5\pi 10^7 t - 9^\circ)\quad (mA).$$

P.6-3 Using phasors with a sine reference:

$$\bar{B}_1 = \bar{a}_\phi \frac{\mu_0 I_1}{2\pi r} \longrightarrow \Phi_{12} = \int_{S_2} \bar{B}_1\cdot d\bar{s}_2 = \frac{\mu_0 I_1 h}{2\pi}\int_d^{d+w}\frac{dr}{r}$$
$$v_2 = -\frac{d\Phi_{12}}{dt} \longrightarrow \text{Phasors: } V_2 = -j\omega\Phi_{12};\qquad = \frac{\mu_0 I_1 h}{2\pi}\ln\left(1+\frac{w}{d}\right).$$
$$I_2 = \frac{V_2}{R+j\omega L} = -\frac{j\omega\mu_0 I_1 h}{2\pi(R+j\omega L)}\ln\left(1+\frac{w}{d}\right)$$
$$= -\frac{\omega\mu_0 I_1 h}{2\pi(\omega L - jR)}\ln\left(1+\frac{w}{d}\right) = -\frac{\omega\mu_0 I_1 h}{2\pi\sqrt{R^2+\omega^2L^2}}\ln\left(1+\frac{w}{d}\right)e^{j\tan^{-1}(R/\omega L)}$$
$$\longrightarrow i_2 = -\frac{\omega\mu_0 I_1 h}{2\pi\sqrt{R^2+\omega^2L^2}}\ln\left(1+\frac{w}{d}\right)\sin\left(\omega t + \tan^{-1}\frac{R}{\omega L}\right).$$

P. 6-4 $\quad \bar{B}_1 = -\bar{a}_x \dfrac{\mu_0 I_0}{2\pi r}.$

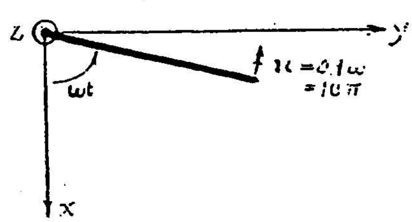
Induced emf in loop $= \oint (\bar{u}_2 \times \bar{B}_1)\cdot d\bar{\ell}_2.$

$$\longrightarrow \mathcal{V}_2 = \frac{\mu_0 I_0 h u_0}{2\pi}\left(\frac{1}{d} - \frac{1}{d+w}\right),$$

in a clockwise direction.

$$i_2 = -\frac{\mathcal{V}_2}{R} = -\frac{\mu_0 I_0 h u_0 w}{2\pi d(d+w)}.$$

P. 6-5



a) If $L=0$:

$$i = \frac{1}{R}(\bar{u}\times\bar{B})\cdot(-\bar{a}_z 0.1)$$
$$= \frac{1}{0.5}(10\pi\times 0.04)\times 0.1\sin\omega t$$
$$= 0.251\sin 100\pi t \quad (A).$$

b) If $L = 0.0035$ (H):

$$\omega L = 100\pi\times 0.0035 = 1.1\ (\Omega),$$
$$\frac{1}{R+j\omega L} = \frac{1}{0.5+j1.1} = \frac{1}{1.208\angle 65.6^\circ}.$$

$$\longrightarrow i_2 = \frac{(10\pi\times 0.04)\times 0.1}{1.208}\sin(\omega t - 65.6^\circ)$$
$$= 0.104\sin(100\pi t - 65.6^\circ)\ (A).$$

P. 6-6 $\quad \Phi = \bar{B}(t)\cdot\bar{S}(t) = -5\cos\omega t \times 0.2\,(0.7-x)$

$$= -\cos\omega t\,[0.7 - 0.35(1-\cos\omega t)]$$
$$= -0.35\cos\omega t\,(1+\cos\omega t) \quad (mT).$$

$$i = -\frac{1}{R}\frac{d\Phi}{dt} = -\frac{1}{R}\,0.35\omega(\sin\omega t + \sin 2\omega t)$$
$$= -1.75\,\omega(\sin\omega t + \sin 2\omega t)$$
$$= -1.75\,\omega\sin\omega t\,(1+2\cos\omega t) \quad (mA).$$

P.6-7 Conduction current density: σE.

Displacement current density: $j\omega D = j\omega\epsilon_0\epsilon_r E$.

For equal magnitude: $\sigma = 2\pi\epsilon_0\epsilon_r f$,

$$\text{or} \quad f = \frac{\sigma}{2\pi(\epsilon_0\epsilon_r)} = 18\times 10^9\left(\frac{\sigma}{\epsilon_r}\right) \quad (Hz).$$

a) Seawater: $f = 18\times10^9\left(\frac{4}{72}\right) = 10^9\,(Hz) = 1\,(GHz)$.

b) Moist soil: $f = 18\times10^9\left(\frac{10^{-3}}{2.5}\right) = 7.2\times10^6\,(Hz)$, or $7.2\,(MHz)$.

P.6-8

a) $$\left|\frac{\text{Displacement current}}{\text{Conduction current}}\right| = \frac{\omega\epsilon}{\sigma} = \frac{(2\pi\times100\times10^9)\times\frac{1}{36\pi}\times10^{-9}}{5.70\times10^7} = 9.75\times10^{-8}.$$

b) In a source-free conductor:

$$\bar{\nabla}\times\bar{H} = \sigma\bar{E}, \qquad (1)$$

$$\bar{\nabla}\times\bar{E} = -j\omega\mu\bar{H}. \qquad (2)$$

$$\bar{\nabla}\times(1):\ \bar{\nabla}\times\bar{\nabla}\times\bar{H} = \bar{\nabla}(\bar{\nabla}\cdot\bar{H}) - \bar{\nabla}^2\bar{H} - \sigma\bar{\nabla}\times\bar{E}. \qquad (3)$$

But $\bar{\nabla}\cdot\bar{H} = 0$, Eq. (3) becomes

$$\bar{\nabla}^2\bar{H} + \sigma\bar{\nabla}\times\bar{E} = 0. \qquad (4)$$

Combining (2) and (4):

$$\bar{\nabla}^2\bar{H} - j\omega\mu\sigma\bar{H} = 0.$$

P.6-9 $\bar{H}_1 = \bar{a}_x30 + \bar{a}_y40 + \bar{a}_z20$.

$B_{2n} = B_{1n} \longrightarrow H_{2z} = \frac{1}{\mu_{r2}}H_{1z} = 10$.

$\longrightarrow B_{2z} = B_{1z} = 20\mu_0$.

$\bar{a}_{n2}\times(\bar{H}_1 - \bar{H}_2) = \bar{J}$

$\longrightarrow \bar{a}_z\times(\bar{a}_x30 + \bar{a}_y40 - \bar{H}_2) = \bar{a}_x5$.

$\longrightarrow H_{2x} = 30,\ H_{2y} = 45$.

a) $\bar{H}_2 = \bar{a}_x30 + \bar{a}_y45 + \bar{a}_z10$ (A/m). b) $\bar{B}_2 = 2\mu_0\bar{H}_2$ (T).

c) $\alpha_1 = \tan^{-1}\frac{\sqrt{30^2+40^2}}{20} = 68.2^\circ$. d) $\alpha_2 = \tan^{-1}\frac{\sqrt{30^2+45^2}}{10} = 79.5^\circ$.

P. 6-10 Medium 1: Free space.
Medium 2: $\mu_2 \to \infty$. H_2 must be zero so that B_2 is not infinite.

Boundary Conditions: $\bar{a}_{n1} \times \bar{H}_1 = \bar{J}_s$, $B_{1n} = B_{2n}$.
$E_{1t} = E_{2t}$, $\bar{a}_{n2} \cdot (\bar{D}_1 - \bar{D}_2) = \rho_s$.

P. 6-13 $E(z,t) = 50 \cos(2\pi 10^9 t - kz)$ (V/m) in air.

$E_0 = 50$ (V/m).
$f = 10^9$ (Hz), $T = \frac{1}{f} = 10^{-9}$ (s).
$\lambda = \frac{c}{f} = \frac{3\times 10^8}{10^9} = 0.3$ (m),
$k = \frac{2\pi}{\lambda} = \frac{20}{3}\pi$.

a) At $z = 100.125\lambda$, $kz = 200.25\pi$, which is same as for $kz = 0.25\pi$, or $\pi/4$.

It is a plot of $E(t) = 50 \cos 2\pi 10^9 (t - T/8)$ (V/m).

b) At $z = -100.125\lambda$, it is a plot of
$$E(t) = 50 \cos 2\pi 10^9 (t + T/8) \quad \text{(V/m)}.$$

c) At $t = T/4$, we plot versus z the following sinusoidal function:
$$E(z, \tfrac{T}{4}) = 50 \cos\left(-kz + \tfrac{\omega T}{4}\right) = 50 \cos\left[-k\left(z - \tfrac{\lambda}{4}\right)\right]$$
$$= 50 \cos \tfrac{20\pi}{3}(z - 0.075) \quad \text{(V/m)}.$$

P. 6-14 Use phasors and cosine reference.

$\bar{E} = \bar{a}_x E_0 e^{j\psi}$; $\bar{E}_1 = \bar{a}_x 0.03 e^{-j\pi/2}$; $\bar{E}_2 = \bar{a}_x 0.04 e^{-j\pi/3}$.
$$\bar{E} = \bar{E}_1 + \bar{E}_2 = \bar{a}_x \left[0.03 e^{-j\pi/2} + 0.04 e^{-j\pi/3}\right]$$
$$= \bar{a}_x \left[-j0.03 + 0.04\left(\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right)\right] = \bar{a}_x 0.068 e^{-j72.8^\circ}$$
$\longrightarrow E_0 \doteq 0.068$ (V/m), $\psi = -72.8^\circ$.